
Omni-Persona: Systematic Benchmarking and Improving Omnimodal Personalization

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Abstract

While multimodal large language models have advanced across text, image, and audio, personalization research has remained primarily vision-language, with unified omnimodal benchmarking that jointly covers text, image, and audio still limited, and lacking the methodological rigor to account for absent-persona scenarios or systematic grounding studies. We introduce Omni-Persona, the first comprehensive benchmark for omnimodal personalization. We formalize the task as cross-modal routing over the *Persona Modality Graph*, encompassing 4 task groups and 18 fine-grained tasks across ~ 750 items. To rigorously diagnose grounding behavior, we propose *Calibrated Accuracy* (Cal), which jointly rewards correct grounding and appropriate abstention, incorporating absent-persona queries within a unified evaluation framework. On our dedicated experiments, three diagnostic findings emerge: (i) open-source models show a consistent audio-vs-visual grounding gap that RLVR partially narrows via dense rule-based supervision; (ii) answerable recall and parameter scale are incomplete diagnostics, since strong recall can coexist with absent-persona hallucination and larger models do not always achieve higher Cal, exposing calibration as a separate evaluation axis; and (iii) SFT is bounded by the difficulty of constructing annotated ground-truth supervision at scale, while RLVR generalizes more consistently through outcome-level verifiable feedback yet drifts toward conservative behavior and lower generation quality under our reward design. Omni-Persona thus serves as a diagnostic framework that surfaces the pitfalls of omnimodal personalization, guiding future post-training and reward design.

1 Introduction

The landscape of large generative models has expanded rapidly toward *omnimodal* systems capable of processing or even generating across text, image, and audio within a single model [1–5]. This convergence of modalities broadens the task scope that a single model can handle and moves the community closer to the vision of a personal AI assistant, one that can recognize a user’s face and voice, recall their biographical context, and ground responses in individual identity.

Despite this momentum, multimodal personalization research has remained primarily focused on vision-language settings [6–9], leaving three key gaps that limit progress toward true omnimodal deployment. First, existing benchmarks have rarely provided unified coverage across all three modalities: while vision and text are well-represented, systematic treatment of audio signals such as voice identity, emotional tone, and conversational context remains limited. Second, real-world retrieval is inherently noisy, often yielding contexts where the queried identity is completely *absent*. Yet, personalization is typically evaluated under *well-controlled settings*, such as explicit identity naming [10, 6, 8] or care-

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fully designed caption-based distractors [9], that assume the target is always present. Consequently, these artificial setups and their recall-only protocols fail to expose this critical failure mode. Third, *realistic personalization scenarios* (for example, identifying a person from a face image or voice clip and then answering a query about that individual) have not been systematically studied. Without a benchmark that addresses all three gaps, the community lacks a principled way to diagnose *when* and *how* current omnimodal models fail at personal grounding. While recent studies [8, 11, 12, 9] each address important aspects of the multimodal personalization problem, substantial gaps remain in audio grounding, absent-persona coverage, and realistic evaluation.

To this end, we introduce *Omni-Persona*, the first evaluation-only benchmark for omnimodal personalization, offering systematic cross-modal coverage with full support for *audio* as a persona modality and *absent-persona* cases. We formalize each user’s multimodal profile through the *Persona Modality Graph* (PMG). In this graph-based abstraction, individual user profiles (comprising a profile image, biographical text, and personal audio) act as context nodes. We frame omnimodal personalization as a cross-modal routing problem: the model must evaluate incoming queries and correctly establish a directed linkage (edge) to the matching context node to ground its response.

Omni-Persona spans 4 task groups and 18 fine-grained tasks over ~ 750 evaluation items, enabling systematic evaluation of both perceptual matching and grounded retrieval. To reflect real-world retrieval imperfections, we explicitly include absent-persona samples, where the ground-truth persona is entirely missing from the retrieved context. This setting introduces retrieval noise and captures a crucial challenge overlooked by prior multimodal personalization benchmarks. Finally, because recall alone cannot capture hallucination and over-abstention, we employ *Calibrated accuracy* (Cal) as our primary metric, equally rewarding correct grounding for answerable items and correct abstention (*i.e.*, forming no edge) for absent-persona items.

Beyond benchmarking, we investigate which post-training regimes best align current omnimodal models with personalization. While previous studies [8, 9] have highlighted the efficacy of RLVR for multimodal personalization in image captioning tasks, we broaden this investigation to omnimodal personalization. Specifically, we rigorously compare supervised fine-tuning (SFT) and reinforcement learning with verifiable rewards (RLVR) to reveal which post-training regime is most suitable, and which specific aspects drive improvements in *omnimodal personalization*. Prior work establishes that SFT is heavily influenced by data quality [13] and scale [14], whereas recent RLVR methods rely on carefully specified verifiable reward signals, such as rule-based accuracy and format rewards [15]. Motivated by this distinction, we conduct SFT on our rigorously curated ground-truth annotation corpora at two different scales (1K and 10K). We contrast this with an RLVR (without SFT warmup) recipe that jointly optimizes perception and retrieval. This RLVR approach utilizes rule-based perceptual verification, alongside LLM-as-a-judge retrieval verification for free-form QA.

Our comparative analysis reveals a distinct trade-off. SFT is constrained by the difficulty of constructing high-quality, in-domain ground-truth supervision for diverse open-ended scenarios, which often prevents broader task coverage from translating into Cal gains. Our RLVR approach mitigates this limitation by using verifiable reward signals to optimize for task-level correctness directly, rather than requiring reference responses for every training instance. However, it introduces a separate trade-off: under a binary reward design, smaller models tend to drift toward over-conservative abstention. We comprehensively validate these findings across both Qwen2.5-Omni and Gemma4 architectures.

Our contributions are as follows:

- (1) **Omni-Persona Benchmark and PMG Formulation.** We introduce *Omni-Persona*, the first comprehensive evaluation-only benchmark for omnimodal personalization. Built on the *Persona Modality Graph* (PMG), it formalizes contextual grounding over retrieved persona evidence and integration of raw-form multimodal contexts, spanning 4 task groups and 18 fine-grained tasks across image, text, and vocal audio.
- (2) **Addressing Recall Blind Spots with Absent-Persona Evaluation.** While prior personalization benchmarks heavily rely on answerable-only recall, we elevate absent-persona queries to a first-class evaluation dimension. By coupling these unanswerable queries with hard distractors and retrieval noise, we propose a calibrated accuracy metric that jointly assesses correct grounding and appropriate abstention. This balanced approach exposes critical hallucination and over-abstention behaviors often masked by recall-only protocols.
- (3) **Diagnostic Analysis of Omnimodal Personalization and Post-Training.** We systematically evaluate closed-source models and open-source models, with post-training analysis conducted on

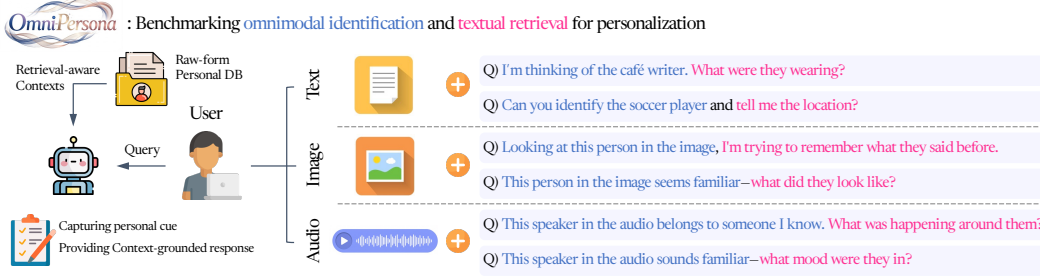


Figure 1: **Formulation of omnimodal personalization in the Omni-Persona benchmark.** As illustrated, a user query consists of a target modality and a text prompt, which we decompose into two main axes: **blue** elements denote the process of perceptual identification, whereas **pink** elements highlight the realistic textual retrieval desired by the user. Note that our underlying assumption is that the raw-form contexts are pre-retrieved; we do not evaluate the retrieval process itself.

the latter. Our analysis reveals a visual-over-audio grounding asymmetry in open-source models and identifies distinct failure modes across SFT and RLVR. Together, these findings provide a model-specific diagnostic map to guide future research on omnimodal personalization.

2 Related Works

Multimodal Personalization Methods. Early personalized vision-language models (VLMs) [10, 6, 16] repurpose off-the-shelf models to recognize user-defined concepts via zero- or few-shot retrieval, yet remain brittle when new concepts must be incorporated dynamically into a user’s memory. Post-training-based approaches subsequently emerged to mitigate this rigidity. Hao et al. [7] first demonstrated that SFT over retrieval-augmented user contexts enables coherent personalized response generation, but its reliance on costly large-scale caption annotations limits practical scalability. To alleviate this annotation burden, Oh et al. [8, 9] introduced RLVR-based methods, validating their utility in multi-concept image captioning [8] and reactive/proactive personalization scenarios [9]. Despite this progress, *audio* has received comparatively limited attention throughout this evolution: visual identity and biographical text [17, 11] have served as the predominant persona modalities, and speaker voice or conversational audio have rarely been integrated within a unified omnimodal personalization framework.

Evaluation Protocols for Multimodal Personalization. The evaluation protocols accompanying these methods, including those of [6, 8–10], rely heavily on recall-centric metrics. Such metrics primarily reward surface-level signals, such as name recall and contextual dialogue snippets, that can be directly reinforced during post-training. As a result, broader generation quality, calibration under absent-persona queries, and the trade-offs introduced by RL-based post-training remain largely unmeasured. This limitation is further compounded by existing benchmarks, which often operate under tightly controlled settings and abstract away realistic retrieval noise. To overcome such limitations, our benchmark unveils failure modes that are otherwise hidden beneath recall-only evaluation in multimodal personalization. Specifically, it exposes hallucination and over-abstention behaviors that conventional recall-centric metrics fail to capture. To the best of our knowledge, no prior work has unified interleaved omnimodal contexts, absent-persona evaluation, and a rigorous diagnostic protocol within a single comprehensive benchmark. Further related work is discussed in Appendix A.

3 Problem Formulation

As illustrated in Figure 1, we formally define *omnimodal personalization*, extending the vision-language personalization paradigm [6–11, 17] to incorporate audio as a persona modality alongside vision and text.

Formal Definition. We formalize omnimodal personalization as follows. Let a user’s personal memory be denoted by $\mathcal{M} = \{(v_i, a_i, t_i)\}_{i=1}^N$, where each entry is a triplet comprising a visual identity v_i , an audio sample a_i , and an associated text descriptor t_i . Specifically, v_i may represent

a profile image or an appearance snapshot, a_i a 5–15 s voice sample or a conversational recording, and t_i dialogue or biographical information. Given a new query comprising a user prompt alongside a textual cue t_q , a visual image v_q , or an audio clip a_q , the relevant entries are retrieved from the memory $\mathcal{M}$ to construct the aggregated top- K context $\{\mathcal{C}_i\}_{i=1}^K$, where $\mathcal{C}_i = (v_i, a_i, t_i)$. Following this retrieval, the model must, at inference time:

1. *Recognize* which specific entry $\mathcal{C}_j$ within the aggregated contexts $\{\mathcal{C}_i\}_{i=1}^K$ corresponds to the provided query cue (v_q , a_q , or t_q); and
2. *Selectively extract and integrate* the specific details pertinent to the query from the associated text t_j of the identified entry $\mathcal{C}_j$ into a contextually grounded response.

The model must first accurately perceive the query and then ground its personalized response in the query-relevant context. Furthermore, the retrieved contexts $\{\mathcal{C}_i\}_{i=1}^K$ arrive in an interleaved format, where the components of each entry $\mathcal{C}_i = (v_i, a_i, t_i)$ appear in an ordered sequence.

Why Raw Omnimodal Context Matters. Previous textual-memory-based multimodal personalization works [11, 17, 18] rely on converting multimodal signals into compact textual descriptions, introducing an inherent *lossy compression* that inevitably discards fine-grained identity information. This information bottleneck is especially problematic for attributes such as voice and visual appearance, where subtle personal traits like vocal timbre and facial geometry cannot be faithfully encoded in text. Consequently, text-only memory falls short in capturing true persona-defining characteristics. To address this limitation, we focus on personalization derived directly from *raw omnimodal context*, grounding the model’s behavior directly in images and audio as perceptual signals.

Research Goal: Strengthening Grounding Expressiveness. We define *expressiveness* in the context of personalization as the extent to which a model can faithfully extract, integrate, and surface personal identity signals from retrieved omnimodal context in its response. The overarching goal of this work is therefore to *define, measure, and systematically improve this grounding expressiveness*.

Scope: Contextual Grounding over Retrieval. We decompose omnimodal personalization into two conceptually distinct sub-problems: (i) *retrieval*, identifying which memories in a user’s history are relevant to a given query, and (ii) *contextual grounding*, integrating retrieved multimodal evidence into a faithfully personalized response. These two components are separable by construction. Accordingly, we decouple the two and focus this work on grounding: given a pre-retrieved omnimodal context, can a model correctly determine which context a query refers to, extract the relevant personal details, and generate a response faithfully grounded in that context? This choice isolates the model’s intrinsic *expressiveness* from retrieval quality.

4 Omni-Persona: Benchmarking Omnimodal Identification and Retrieval

We instantiate Omni-Persona through the Persona Modality Graph (PMG), where each node is defined as a triplet representing an individual’s omnimodal data. In this framework, personalization scenarios are modeled by the interconnections established between these nodes. Building upon this formulation, we propose a novel benchmark that simulates realistic personalization challenges, specifically focusing on modality matching (*i.e.*, graph linkage) within the PMG.

Persona Modality Graph (PMG) and Task Formulation.

We formalize omnimodal personalization as a cross-modal routing problem over a PMG, $\mathcal{G} = (\mathcal{V}, \mathcal{E})$. The vertices $\mathcal{V}$ consist of a query node $\mathcal{Q}$ and retrieved context nodes $\mathcal{C}_1, \dots, \mathcal{C}_K$, where each node can encompass visual (v), audio (a), and textual (t) modalities, as represented in Figure 2.

The core task is to determine whether a retrieved context contains the target persona and to establish a directed linkage (edge) $e_{q \rightarrow j} \in \mathcal{E}$ accordingly. Based on the provided query modality, we categorize the routing process into four primary matching scenarios:

- (1) *Image-to-Image (I2I)*: matching visual identity to an image query (*i.e.*, visual identification);
- (2) *Audio-to-Audio (A2A)*: matching voice identity to an audio query (*i.e.*, voice identification);
- (3) *Text-to-Text (T2T)*: matching textual attributes to a text query (*i.e.*, same-modal semantic); and

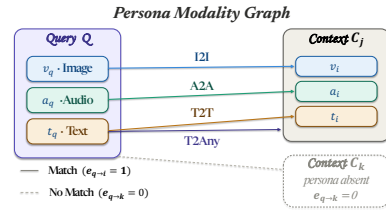


Figure 2: PMG illustration.

Table 1: Omni-Persona evaluation scenario groups. Each group tests whether a model can retrieve the correct persona information from a query in a different modality-matching setting.

Group	Setting	What the Model Must Match	Core Challenge
1: I2I	Image query	Match a person’s face to the same visual identity in the persona context	Visual identity recognition
2: A2A	Audio query	Match a person’s voice to the same speaker identity in the persona context	Voice identity recognition
3: T2T	Text query	Match a textual cue to the relevant textual persona attribute	Same-modal semantic retrieval
4: T2Any	Text query	Match the meaning of a text cue to relevant text, image, or audio information (e.g., visual or emotional description)	Cross-modal semantic grounding

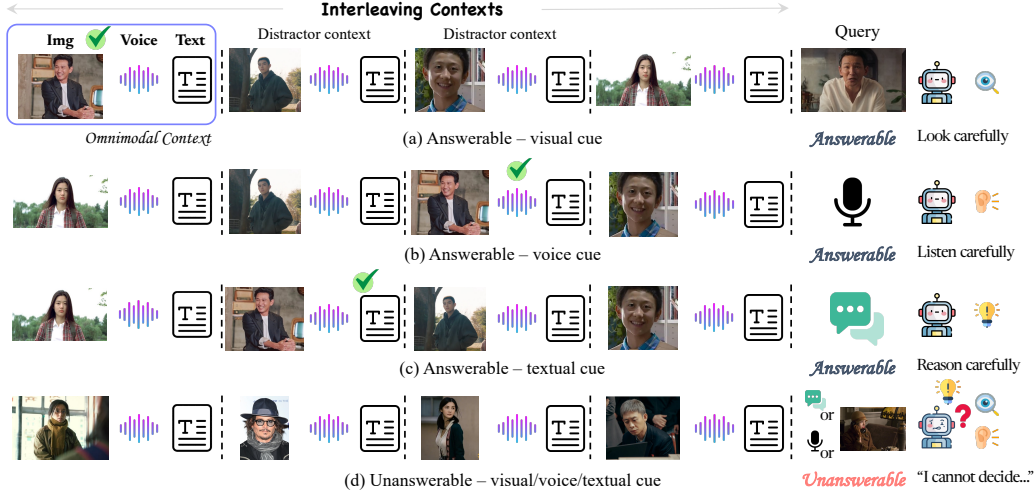


Figure 3: Qualitative examples of context construction, personalization cues, distractors, and unanswerable cases in Omni-Persona.

- (4) *Text-to-Any (T2Any)*: aligning the semantic meaning of a text query with the cross-modal content of text, image, or audio (*i.e.*, cross-modal semantic).

Crucially, this formulation natively handles absent-persona calibration. If a context $\mathcal{C}_j$ contains the target persona, an active edge is formed ($e_{q \rightarrow j} = 1$), allowing the model to traverse the graph to extract and integrate grounded details from the associated text. Conversely, if the target persona is entirely absent from the provided contexts, no edge is formed ($e_{q \rightarrow j} = 0$), requiring the model to confidently abstain. This unified framework systematically yields the 4 scenario groups in Table 1 and 18 fine-grained tasks detailed in Appendix H.

Benchmark Design Principles. Designed around natural, human-centric interaction scenarios, Omni-Persona is, to our knowledge, the first personalization benchmark to incorporate *audio* as a full persona modality alongside image and text, and to systematically incorporate *unanswerable* items, where the queried persona is absent from the retrieved context, as a primary evaluation dimension. Unlike prior personalization benchmarks [6, 8, 9] that measure only recall (whether the model retrieves the correct persona when it is present), Omni-Persona jointly evaluates *grounding recall* and *abstention*, reflecting the dual challenge of real-world retrieval systems where the queried person may not be in the retrieved contexts at all. Furthermore, cross-modal task design, which requires the model to bridge audio evidence to visual descriptions or vice versa, enables measurement of per-modality grounding bias that unimodal tasks cannot reveal.

Robustness Under Retrieval Imperfection. Because real retrieval pipelines are noisy, Omni-Persona explicitly introduces two classes of perturbation into the evaluation benchmark. The first, *hard distractors*, involves context entries from individuals who share visual or vocal similarities with

Table 2: **Systematic Omni-Persona benchmark results.** Complementary metrics include 1-FalseAbs and TrueAbs. FA, TA each stands for False Abstention, True Abstention, respectively. **Green** indicates improvement and **Red** indicates regression compared to the base model after RLVR.

Model	Overall Ans	Overall Cal	I2I			A2A			T2T			T2Any			Add. Metrics (×100)			
			Ans	Unans	Cal	Ans	Unans	Cal	Ans	Unans	Cal	Ans	Unans	Cal	1-FA	TA	Avg	
Closed-source Models																		
Gemini-3.1-Pro	69.8	76.7	67.0	93.1	80.0	61.8	97.1	79.4	74.7	41.5	58.1	75.9	69.2	72.6	71.4	83.6	77.5	
Gemini-3-Flash	71.4	45.7	68.7	14.7	41.7	66.4	27.7	47.1	77.1	24.4	50.7	73.5	13.8	43.7	95.9	20.0	58.0	
Gemini-3.1-Flash-lite	52.8	42.0	39.1	14.7	26.9	44.5	36.5	40.5	71.1	36.6	53.8	56.6	55.4	56.0	93.9	31.2	62.6	
Open-source Models																		
MiniCPM-o 4.5 (Think)	51.8	33.6	38.3	17.2	27.8	25.5	18.2	21.9	75.9	24.4	50.1	67.5	16.9	42.2	94.6	15.4	55.0	
Phi-4 Multimodal	52.5	40.4	44.3	31.0	37.7	24.5	37.2	30.9	69.9	19.5	44.7	71.1	24.6	47.8	88.0	28.3	58.2	
Qwen-Omni series	Qwen2.5-Omni-3B	49.3	43.6	43.5	44.0	43.7	30.9	32.1	31.5	63.9	53.7	58.8	59.0	38.5	48.7	75.2	37.9	56.6
	+ SFT (1K)	52.4	45.2	41.7	43.1	42.4	35.5	31.4	33.4	68.7	53.7	61.2	63.9	44.6	54.2	74.4	38.0	56.2
	+ SFT (10K)	45.6	41.6	36.5	44.0	40.2	29.1	32.1	30.6	60.2	53.7	56.9	56.6	38.5	47.5	75.2	37.6	56.4
	+ RLVR	54.7	55.2	49.6	59.5	54.5	43.6	45.3	44.4	71.1	70.7	70.9	60.2	61.5	60.9	56.8	55.7	56.2
	Δ vs. Base (RLVR)	+5.4	+11.6	+6.1	+15.5	+10.8	+12.7	+13.2	+12.9	+7.2	+17.0	+12.1	+1.2	+23.0	+12.2	-18.4	+17.8	-0.4
	Qwen2.5-Omni-7B	47.9	34.2	39.1	26.7	32.9	28.2	13.1	20.7	61.4	48.8	55.1	62.7	18.5	40.6	83.6	20.5	52.1
	+ SFT (1K)	47.2	34.3	36.5	28.4	32.5	31.8	13.1	22.5	59.0	46.3	52.7	61.4	21.5	41.5	84.4	21.4	52.9
	+ SFT (10K)	45.9	33.0	40.0	26.7	33.4	25.5	13.1	19.3	55.4	48.8	52.1	62.7	15.4	39.0	83.9	20.1	52.0
	+ RLVR	48.3	38.0	42.6	31.9	37.3	27.3	18.2	22.8	66.3	56.1	61.2	66.3	21.5	43.9	78.5	27.6	53.1
	Δ vs. Base (RLVR)	+0.4	+3.8	+3.5	+5.2	+4.4	-0.9	+5.1	+2.1	+4.9	+7.3	+6.1	+3.6	+3.0	+3.3	-5.1	+7.1	+1.0
Gemma4-series	Qwen3-Omni-30B	49.1	31.5	44.3	22.4	33.4	20.9	9.5	15.2	68.7	17.1	42.9	62.7	18.5	40.6	92.8	16.2	54.5
	Gemma4-E2B	46.6	36.4	45.2	17.2	31.2	21.8	56.2	39.0	57.8	2.4	30.1	61.4	4.6	33.0	89.0	26.2	57.6
	+ SFT (1K)	45.7	35.7	43.5	16.4	29.9	16.4	56.2	36.3	59.0	2.4	30.7	63.9	4.6	34.2	88.8	25.7	57.3
	+ SFT (10K)	48.3	36.9	42.6	17.2	29.9	22.7	54.7	38.7	61.4	0.0	30.7	66.3	7.7	37.0	88.5	25.5	57.0
	+ RLVR	47.8	42.4	43.5	29.3	36.4	26.4	64.2	45.3	62.7	4.9	33.8	67.5	13.8	40.7	80.6	37.0	58.8
	Δ vs. Base (RLVR)	+1.2	+6.0	-1.7	+12.1	+5.2	+4.6	+8.0	+6.3	+4.9	+2.5	+3.7	+6.1	+9.2	+7.7	-8.4	+10.8	+1.2
	Gemma4-E4B	65.3	52.6	65.2	37.9	51.6	41.8	67.9	54.9	74.7	4.9	39.8	79.5	15.4	47.5	77.8	39.9	58.9
	+ SFT (1K)	65.3	51.6	67.8	22.4	45.1	35.5	75.9	55.7	73.5	2.4	38.0	84.3	18.5	51.4	80.3	37.9	59.1
	+ SFT (10K)	66.2	53.7	67.8	37.9	52.9	42.7	72.3	57.5	79.5	2.4	41.0	74.7	13.8	44.3	78.5	41.2	59.9
	+ RLVR	68.8	62.0	67.0	44.8	55.9	58.2	91.2	74.7	75.9	9.8	42.8	78.3	26.2	52.2	74.7	55.2	65.0
Δ vs. Base (RLVR)	+3.5	+9.4	+1.8	+6.9	+4.3	+16.4	+23.3	+19.8	+1.2	+4.9	+3.0	-1.2	+10.8	+4.7	-3.1	+15.3	+6.1	

the target. The second, *no-GT retrieval*, entirely omits the ground-truth persona from the context, demanding structured abstention instead of hallucinated matching. This rigorous setup guarantees a comprehensive evaluation across diverse omnimodal tasks. With approximately 50% of the evaluation samples being no-GT, the benchmark systematically probes the model’s resistance to hallucination, an essential desideratum when integrating with RAG systems [19–21].

5 Experiments

Our training study investigates which post-training regime most effectively aligns current omnimodal models for personalization. To this end, we systematically evaluate diverse models on our benchmark, elucidate the underlying behaviors surfaced by our evaluation metrics, and conduct an in-depth model debugging analysis to identify what is fundamentally required to advance omnimodal personalization. Due to space limitations, exhaustive details on data curation and implementation for the post-training experiments are deferred to Appendix D.

5.1 Experimental Setup

Used Models. We evaluate four open-source omnimodal backbones (Gemma4-E2B-it, Gemma4-E4B-it, Qwen2.5-Omni-3B, and Qwen2.5-Omni-7B) [1, 4] under four training regimes: zero-shot, SFT-1K, SFT-10K, and RLVR. Within the Gemma4 series, audio processing is supported exclusively by the E2B and E4B variants. As an upper-bound reference, we additionally include the closed-source Gemini-3 family [3], together with three open-source baselines: Qwen3-Omni-30B-A3B-Instruct [22], Phi-4-multimodal-Instruct [23], and MiniCPM-o 4.5 (thinking) [2]. All post-training is performed with LoRA [24], using ms-swift [25] for SFT and TRL[‡] for RLVR. Full implementation details and user prompt templates are provided in Appendix D and Appendix I, respectively.

SFT Training Setup. We construct a 10K-sample SFT dataset spanning 12 distinct task types, complemented by a 1K subset for efficient ablation studies. This corpus encompasses foundational grounding, audio-centric scenarios, and absent-persona cases designed to promote calibrated abstention. Crucially, we curate this dataset for broad modality alignment across image, audio, and text, rather than narrow, benchmark-specific optimization. We emphasize that constructing a training

[‡]<https://github.com/huggingface/trl>

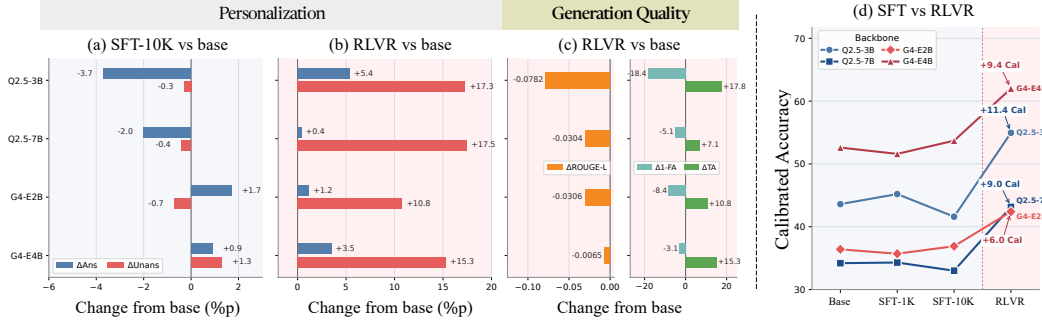


Figure 4: **Performance across post-training regimes.** Personalization: (a) SFT scaling does not directly translate to gains in open-ended personalization scenarios. (b) RLVR improves recall and unanswerable performance. Generation quality: (c) RLVR reduces both ROUGE-L and 1 – FA compared with the base model. (d) RLVR yields stronger calibrated accuracy gains than SFT scaling.

corpus for SFT is fundamentally constrained by several factors: (i) the *inherent noise* in synthesizing high-quality ground truth responses for diverse personalization scenarios [7, 8]; (ii) the *unpredictability* of the test-time query distribution [8, 26, 17]; and (iii) the *scarcity* of large-scale, paired real-world multimodal data [27, 28], necessitating a reliance on synthetic samples that may introduce domain bias. These limitations collectively make SFT alone insufficient for ensuring predictable personalization coverage at test time, consistent with recent omnimodal post-training studies showing that RL-based objectives can substantially outperform SFT under matched data and compute budgets [29–31]. Consequently, we treat SFT as a comparative baseline for RLVR to rigorously analyze how the two post-training regimes shape performance trends.

RLVR Training Setup. To ensure a fair comparison, we perform RLVR on synthetic persona contexts produced by the same pipeline as the SFT corpus. Thus, both regimes use the same type of image–audio–text context triplets and queries. Unlike SFT, however, RLVR does not require reference GT responses. Instead, the model is trained with verifiable binary feedback (*i.e.*, 1 or 0) that checks whether its response satisfies the intended capability, such as perceptual matching or grounded retrieval. We use two complementary reward components:

- (1) *Perception (Rule-based)*: Evaluates visual or auditory persona matching by comparing the model’s binary decision (*e.g.*, yes/no) with the GT label derived from the query–context pairing.
- (2) *Retrieval (LLM-as-a-Judge)*: Evaluates whether the response is grounded in retrieved persona evidence. When the target persona is present, the judge verifies factual support against the GT answer and context; when absent, it checks whether the model correctly abstains.

For retrieval reward, we use GPT–5.4 to generate training queries and GT answers from the benchmark scenarios, rather than full reference responses, while enforcing a disjoint split to prevent evaluation overlap (see Figure S.1 for an example of the RLVR training framework).

Evaluation Metrics. We treat visual and auditory perception jointly: a model that “*sees, hears, and reasons well*” must *consistently match identical entities and distinguish distinct ones across both modalities*. To rigorously evaluate these capabilities, a faithful assessment of omnimodal personalization must simultaneously account for two critical failure modes that standard recall metrics conflate: *hallucinating an identity absent from the context* and *wrongly answering when the identity is absent*. Accordingly, we evaluate models on a fixed pool of 750 queries, balanced between answerable ($N_{\text{Ans}} = 391$) and unanswerable ($N_{\text{Unans}} = 359$) scenarios. Further judge-reliability analyses are detailed in Appendix C.2. Our evaluation relies on the following key metric groups:

- *Calibrated Accuracy (Cal)*: Our primary metric is defined as $\text{Cal} = \frac{1}{2}(\text{Ans} + \text{Unans})$. *Answerable Recall (Ans)* is evaluated for binary correctness using an LLM-as-a-judge (GPT–5.4–mini), while *Unanswerable Recall (Unans)* is measured via abstention-keyword matching [32]. Cal serves to operationalize a model’s true expressiveness and grounding in omnimodal personalization.
- *Anti-Hallucination Accuracy*: To prevent models from artificially inflating Cal through blanket abstention, we rigorously assess their anti-hallucination reliability. Specifically, we report two complementary metrics: the answerable-side complement of False Abstention ($1 - \text{FA}$), and

the True Abstention (TA) rate on unanswerable items. These metrics capture generation quality, complementing Cal, and higher values indicate better performance across all reported metrics.

5.2 Main Results

Table 2 reports Cal scores across all models and benchmark scenarios. Building on these results, Figure 4 compares post-training regimes along two axes: personalization and generation quality. We additionally report ROUGE-L on answerable items, a traditional metric that avoids the non-determinism of LLM-as-a-judge evaluation.

Closed-Source Models. Cal score exposes a sharp divide that recall alone hides. Gemini-3.1-Pro is the only model that achieves both strong grounding and reliable abstention, leading in Cal at 76.7%. In contrast, Gemini-3-Flash leads in Ans (71.4%) but collapses on Unans, yielding only 45.7% Cal. This low Cal mainly reflects hallucination on absent-persona items, evidenced by its low TA score.

Open-Source Baselines. Among open-source models, parameter scale alone does not guarantee high Cal in our benchmark: Qwen3-Omni-30B *underperforms* Qwen2.5-Omni-3B on Cal (31.5% vs. 43.6%), and MiniCPM-o-4.5 (thinking) and Phi-4 Multimodal similarly trail the 3B Qwen baseline despite generating substantially longer responses. In contrast, Gemma4-E4B emerges as the strongest open-source family, motivating its selection as our primary analysis target for RLVR. We further discuss the discrepancy between scaling and Cal score in Section 6 (Key Finding 2).

Scaling SFT Fails to Bridge the Distribution Gap in Open-Ended Personalization. Expanding the SFT dataset from 1K to 10K samples fails to yield consistent Cal improvements and even leads to performance degradation (Figure 4a,d) for the Qwen series. While the augmented corpus was curated to strengthen omnimodal matching across diverse contexts, this broader coverage does not translate into superior benchmark performance. This discrepancy highlights the rigorous complexity of the open-ended personalization required at test time and provides empirical evidence for the inherent limitations of SFT data construction: simply scaling training volume for general modality alignment is insufficient to bridge the distributional gap in complex, personalized reasoning tasks.

RLVR Delivers Calibration Gains through Verifiable Supervision. Conversely, RLVR consistently enhances Cal across all configurations (Figure 4b). Gemma4-E4B exhibits the largest gain (+9.4 Cal), driven primarily by a marked increase in Unans accuracy, while showing the smallest degradation in the 1 – FA score tradeoff among RLVR models. Despite its compact 4.5B-parameter size, this RL-optimized model surpasses Gemini-3-Flash and establishes a new state of the art among the evaluated open-source models in both Cal and the average of the anti-hallucination metrics (1 – FA and TA), as shown in Table 2. The effectiveness of RLVR stems from its ability to directly reinforce core capabilities, namely perception and retrieval, through dense outcome-based verifiable rewards. This training signal bypasses the need for ground-truth response alignment, which remains a critical bottleneck for SFT in open-ended personalization settings.

The Over-Conservatism Trade-off of RLVR. However, our anti-hallucination accuracies reveal a calibration trade-off in RLVR. As shown in Figure 4c, although RLVR improves TA on unanswerable items for the Qwen2.5-Omni and Gemma4 series, their 1 – FA scores drop below the baseline, indicating more false abstentions on answerable cases. We interpret this as a *reward-induced abstention bias*. Under our binary yes/no reward, abstention is directly rewarded on absent-persona cases, while grounding requires correctly identifying the relevant persona and retrieving the right evidence. Thus, when the model is uncertain, abstention can become the safer action even for answerable inputs, because it avoids the risk of producing an unsupported grounded response. This effect is especially visible in smaller models (Qwen2.5-Omni 3B and Gemma4-E2B 2.3B), where perceptual matching and retrieval are less reliable. More refined reward design, such as asymmetric weighting between grounding and abstention errors, may reduce this effect; we leave such ablations to future work.

6 Towards Omnimodal Personalization: In-depth Model Debugging Analysis

Key Finding 1: Audio Grounding Lags Behind Visual Grounding

Open-source models exhibit a consistent performance deficit in audio grounding compared to visual grounding, resulting in a 15-25%p answerable-recall gap between A2A and I2I tasks. RLVR partially narrows this disparity by providing dense rule-based supervision.

When evaluated independently on visual and auditory tasks, open-source models demonstrate a significantly stronger capability to process visual cues than audio signals. This suggests that authentic audio perception remains a fundamental weakness even in state-of-the-art models, resulting in a pronounced gap between I2I and A2A performance. Notably, the Gemini-3 family stands as the sole exception, where this perceptual gap is not clearly evident. Significantly, by providing explicit perception supervision via rule-based rewards, our RLVR approach narrows this performance gap across all Gemma4 backbones. Consequently, we pose that this perceptual imbalance, often attributed to inherent limitations in representational capacity, can be effectively mitigated by RLVR.

💡 Key Finding 2: Recall and Model Scale Miss the Calibration Axis

Answerable recall and model scale are useful but incomplete diagnostics for personalization. A model with strong recall may still hallucinate when the target persona is absent (low TA), or over-abstain even when the relevant persona is present (low $1 - FA$). Likewise, Cal exposes this missing axis by jointly measuring grounding on answerable cases and abstention on unanswerable cases.

A closer look at Qwen3-Omni-30B illustrates this point. While the 30B model shows superiority in answerable recall for text-grounded tasks, it exhibits performance declines in perceptual matching (I2I, A2A) and unanswerable cases compared to smaller Qwen2.5-Omni variants, resulting in lower overall Cal and the lowest TA score among the Qwen models, which indicates higher hallucination. Similarly, Gemini-3-Flash achieves strong answerable recall but substantially underperforms Gemini-3.1-Pro in calibrated accuracy and TA. These cases demonstrate that omnimodal personalization must account for a broad spectrum of capabilities: perceptual grounding, textual retrieval, and calibrated abstention. Recall-only comparisons often obscure these discrepancies, whereas Cal and anti-hallucination accuracy effectively expose them.

💡 Key Finding 3: SFT and RLVR Reveal Different Post-training Trade-offs

SFT and RLVR fail in different ways. SFT is limited by the difficulty of constructing annotated GT supervision at scale, making it challenging to consistently improve evaluation performance. In contrast, RLVR can target perceptual grounding, textual retrieval, and abstention through verifiable reward signals, even without SFT warmup. However, without carefully designed rewards, RLVR may introduce over-conservative abstention (low $1 - FA$).

SFT is limited by the difficulty of constructing high-quality supervision data at scale. Scaling the data from 1K to 10K does not reliably improve Cal, suggesting that a broader data mixture does not directly translate into better benchmark performance in open-ended personalization scenarios. This can be interpreted as a training-evaluation mismatch [33]. Conversely, RLVR avoids this bottleneck by replacing reference-response imitation with outcome-level supervision. Perception reward reinforces binary visual/audio matching, while retrieval reward reinforces responses grounded in textual evidence. This enables RLVR to improve calibration, especially for Gemma4-E4B, while bypassing the need for GT responses. However, because our binary rewards assign equal weight to correct abstention and correct grounding, models may converge to a lower-risk policy of abstaining too often, improving TA at the cost of $1 - FA$. Overall, these analyses reaffirm the potential of RLVR-based frameworks [8, 9], while emphasizing that future strategies should regularize against reward-hacking behaviors such as using abstention as a shortcut to high rewards.

7 Conclusion

We introduce *Omni-Persona*, the first comprehensive benchmark for omnimodal personalization. By formalizing contextual grounding over retrieved persona evidence and the integration of raw-form omnimodal context, it enables systematic analysis of personalized expressiveness, treats audio as a key persona modality alongside images and text, and adopts absent-persona queries as a core evaluation dimension. We further propose calibrated accuracy and anti-hallucination accuracy, showing that recall-only metrics can obscure hallucination under retrieval noise. Extensive benchmarking shows that, on visual and auditory tasks evaluated independently, open-source models process visual cues substantially more reliably than audio; scaling SFT does not reliably improve performance, reflecting the difficulty of constructing data aligned with open-ended personalization; and RLVR improves

calibration more consistently, though it can induce over-conservative abstention in smaller models and degrade generation quality. Overall, Omni-Persona provides a realistic diagnostic framework for analyzing the strengths and failure modes of omnimodal personalization.

Limitations. Our benchmark uses synthetic audio and text with rigorous model-based filtering, leaving further human verification as future refinement. Free-form LLM-as-a-judge evaluation may introduce residual bias; see Appendix E for further discussion.

8 Acknowledgements

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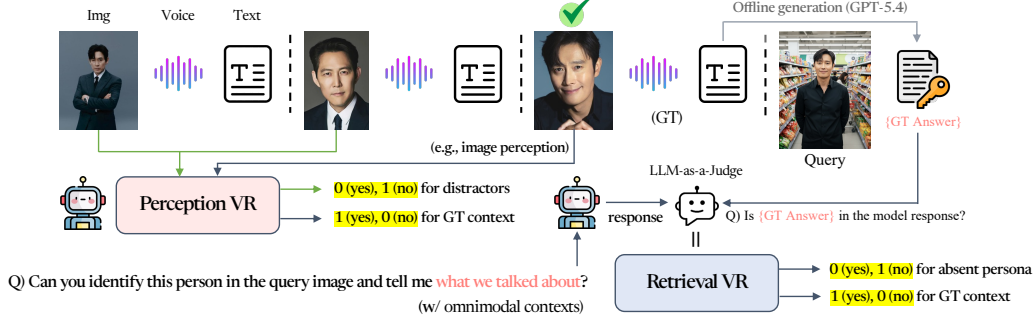


Figure S.1: **Visual representation of the proposed perception and retrieval VR for RLVR.** Note that we strictly utilize binary VR signals. The perception VR provides supervision for correctly identifying the GT context against distractors. Meanwhile, the retrieval VR accounts for supervision in absent-persona scenarios and grounding on the GT answer. Specifically, the GT answer corresponding to the user prompt is generated offline, and an LLM-as-a-judge (*i.e.*, GPT-5.4-mini) is employed to verify its inclusion in the model’s response. Further details of the used data are in Table S.10.

Table S.1: Comparison of previous personalization benchmark against real-world scenarios.

Dataset	Context Nature	Acquisition	Scenarios	Tasks	Modalities	Cost	Evaluation
CoViP [9]	Homogeneous	Synthetic only	Text dialogue	3	I, T	Low (API)	Standard (MCQA)
Omni-Persona (Ours)	Heterogeneous (distractors, no GT)	Mixed (Real & Syn.)	Dialogue, biography	18	I, T, A	Moderate	Advanced (LLM Judge)
Real-world	Noisy & incomplete	Manual (Privacy risks)	Emails, logs, histories	$\gg 18$	I, T, A, V	Prohibitive	Complex (Multi-hop)

A Further Related Works

LLM Personalization. Recent text-only personalization benchmarks evaluate how well LLMs profile and respond to users. For instance, Jiang et al. [26] focus on dynamic profiling, Kim et al. [34] use real user interaction logs for alignment, and Jiang et al. [35] infer latent traits from conversational history. While foundational, these works remain confined to the textual modality, lacking the detailed analyses for multimodal grounding necessary for omnimodal personal assistants.

Comparison with CoViP. Table S.1 contrasts Omni-Persona with CoViP [9], its closest predecessor in the multimodal personalization benchmark space. Relative to CoViP, Omni-Persona: (i) incorporates audio as a persona modality alongside image and text; (ii) incorporates ground-truth-absent queries under multi-distractor contexts as a core evaluation axis; and (iii) broadens the task scope from image captioning to open-ended QA and cross-modal identity matching. Critically, CoViP’s answerable-only multi-choice question answering (MCQA) protocol cannot surface model overconfidence or abstention collapse. By exposing these failure modes through a dual-axis design and extending absent-persona evaluation to audio and cross-modal scenarios, Omni-Persona offers a comprehensive diagnostic framework for omnimodal personalization.

B Preliminaries

Omnimodal and Multimodal Foundation Models. Omnimodal models extend MLLMs toward any-to-any generation across text, vision, and audio. Proprietary systems such as GPT-4o [5] and Gemini-3 [36] have shown strong cross-modal understanding and real-time dialogue capabilities. Among open-source omnimodal models, Qwen2.5-Omni [1] adopts a Thinker–Talker architecture: the Thinker performs multimodal understanding and textual reasoning, while the Talker generates speech responses. MiniCPM-o [2] unifies vision, speech, and language within an end-to-end framework, employing a time-division multiplexing mechanism that interleaves parallel omni-modal streams into periodic time slices, thereby supporting real-time bidirectional speech interaction. Phi-4-

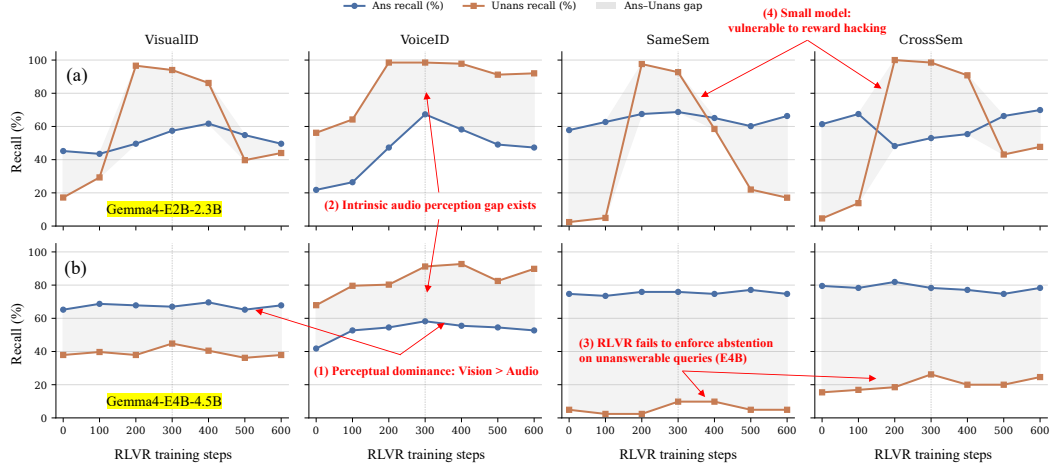


Figure S.2: **Training dynamics during RLVR.** The models exhibit distinct behavioral trajectories throughout the training process. Note that the systematic debugging analysis is visually incorporated within the figure.

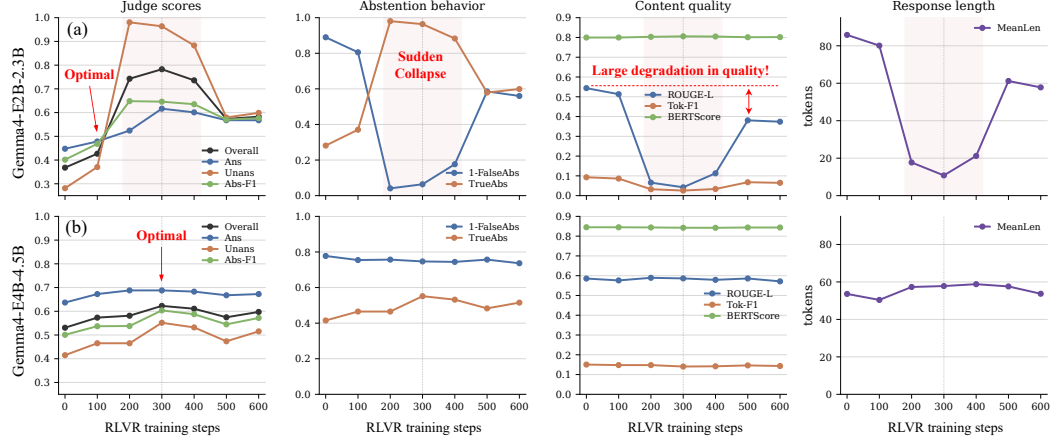


Figure S.3: **Selection of the optimal checkpoint.** While extended training yields a partial recovery from the initial collapse, we select a very early checkpoint to strike a proper balance: improving perceptual enhancement without degrading the underlying generation quality. Note that the exact optimal step may vary depending on the specific on-policy RLVR algorithm employed.

Multimodal [23] integrates text, vision, and speech/audio into a single 5.6B-parameter model through a Mixture-of-LoRAs design, in which modality-specific LoRA adapters and routers are attached to a frozen Phi-4-Mini backbone, enabling competitive multimodal reasoning without cross-modal interference while fully preserving the base language capability.

We further use Gemma4 E-series models (E2B and E4B) [4] as our main analysis models. Gemma4 is not a native omnimodal model, but an MLLM designed for constrained DRAM and flash-memory environments through Per-Layer Embeddings and Grouped-Query Attention, which reduce KV-cache pressure. This makes it a useful on-device-oriented counterpart to larger server-class omnimodal systems, as the strongest zero-shot open-source baseline on Cal (see Tables 1, 2), enabling focused analysis of post-training dynamics.

Table S.2: **Complementary lexical and semantic metric results.** Note: BS = BERTScore, 1-FA = 1-FalseAbs, TA = TrueAbs, MLen = Mean Length.

Model	Accuracy			Generation Quality				Avg	Add. Metrics (×100)			MLen
	Overall(Cal)	Ans	Unans	Abs-F1	ROUGE-L	Tok-F1	BS		1-FA	TA	Avg	
Closed-source Models												
Gemini-3.1-Pro	76.7	69.8	83.6	78.3	60.2	14.6	84.3	66.8	71.4	83.6	77.5	56.5
Gemini-3-Flash	45.7	71.4	20.0	33.0	72.1	20.8	85.9	50.0	95.9	20.0	58.0	43.6
Gemini-3.1-Flash-lite	42.0	52.8	31.2	47.1	56.3	13.5	84.4	46.8	93.9	31.2	62.6	52.9
Open-source Models												
Phi-4 Multimodal	40.4	52.5	28.3	42.9	59.7	13.3	83.9	45.9	88.0	28.3	58.2	89.6
MiniCPM-o (Think)	33.6	51.8	15.4	29.6	64.3	5.7	80.0	40.2	94.6	15.4	55.0	455.7
Qwen3-Omni-30B	32.1	46.8	16.2	26.1	56.1	12.4	84.0	39.1	92.8	16.2	54.5	59.8
Qwen2.5-Omni-3B	43.6	49.3	37.9	47.5	38.8	16.5	85.5	45.6	75.2	37.9	56.6	18.3
SFT (1K)	45.2	52.4	38.0	47.8	38.9	16.9	85.5	46.4	74.4	38.0	56.2	18.3
SFT (10K)	41.6	45.6	37.6	47.5	37.6	16.4	80.3	43.8	75.2	37.6	56.4	18.3
RLVR	55.2	54.7	55.7	55.0	31.0	13.0	85.0	49.9	56.8	55.7	56.2	13.5
Qwen2.5-Omni-7B	34.2	47.9	20.5	32.1	45.9	14.8	84.9	40.1	83.6	20.5	52.1	35.0
SFT (1K)	34.3	47.2	21.4	33.3	46.5	15.0	85.0	40.5	84.4	21.4	52.9	35.0
SFT (10K)	33.0	45.9	20.1	31.9	45.7	15.0	80.0	38.9	83.9	20.1	52.0	35.1
RLVR	38.4	48.3	27.6	36.5	42.8	14.0	84.9	41.8	78.5	27.6	53.1	32.3
Gemma4-E2B	36.4	46.6	26.2	40.2	54.4	9.3	80.0	41.9	89.0	26.2	57.6	85.8
SFT (1K)	35.7	45.7	25.7	39.8	54.0	9.2	79.8	41.5	88.8	25.7	57.3	85.4
SFT (10K)	36.9	48.3	25.5	39.7	54.1	9.3	80.1	42.1	88.5	25.5	57.0	83.5
RLVR	42.4	47.8	37.0	46.8	51.3	8.6	80.0	44.9	80.6	37.0	58.8	80.1
Gemma4-E4B	52.6	65.3	39.9	50.1	58.6	15.1	84.5	52.4	77.8	39.9	58.9	53.6
SFT (1K)	51.6	65.3	37.9	49.4	58.9	16.1	84.9	52.1	80.3	37.9	59.1	47.7
SFT (10K)	53.7	66.2	41.2	51.3	58.7	14.9	84.3	53.0	78.5	41.2	59.9	59.3
RLVR	62.0	68.8	55.2	60.4	58.6	14.1	84.3	57.7	74.7	55.2	65.0	57.8

C Additional Analysis

C.1 RLVR Training Dynamics: Gemma4 E2B vs. E4B

We analyze the step-wise training trajectories of Gemma4-E2B (2.3B) and Gemma4-E4B (4.5B), both trained under identical GSPO configurations across six checkpoints (steps 100 through 600). Comprehensive results are presented in Tables S.3 and S.4. Please refer to the elucidations of additional metrics used in Table S.5, and the additional results in Table S.2.

Gemma4-E4B Maintains Stable Trajectories. The E4B trajectory remains remarkably stable across training (Figure S.2(b)): content metrics fluctuate minimally and abstention rates stay highly consistent. We attribute this stability to the 4.5B-parameter backbone, whose stronger linguistic priors absorb on-policy updates without degenerating into blanket refusal behaviors observed at smaller scales.

Modality Bias and Limitations of Post-Training. The Gemma4 models exhibit an intrinsic perceptual dominance of visual over audio signals, reflecting a bias inherent to the pre-trained backbone (Figure S.2). In the E4B trajectory, answerable recall peaks at roughly 60% as training progresses, demonstrating enhanced audio processing capability following RLVR. The concurrent rise in both answerable and unanswerable recall suggests the model acquires fine-grained audio perception during training.

In contrast, the smaller E2B model shows a sharper gain in unanswerable recall but a steep drop in response length, suggesting over-conservative refusal under the equal-weight binary reward. We view this as a reward-design artifact, where a smaller-capacity policy may choose abstention as the lower-risk action under uncertainty. More refined reward designs may reduce this behavior, which we leave to future work. For E4B, RLVR still struggles to enforce calibrated abstention on the more complex T2T and T2Any categories, suggesting that cross-modal semantic hallucinations cannot be fully suppressed by the current RLVR recipe alone.

In summary, the key takeaways are as follows:

1. **Composite evaluation is essential.** Relying solely on the highest numerical Cal score can be misleading: small-capacity policies can drift toward universal abstention under equal-weight binary rewards. Joint reporting of Ans, Unans, TA, and FA, together with behavioral monitoring along the training trajectory, is necessary to distinguish genuine calibration.

Table S.3: Omni-Persona benchmark ablation results. **Ans** = answerable recall; **Unans** = unanswerable recall; **Cal** = balanced accuracy ($= \frac{\text{Ans} + \text{Unans}}{2}$).

Model	Overall Ans	Overall Cal	I2I			A2A			T2T			T2Any			
			Ans	Unans	Cal	Ans	Unans	Cal	Ans	Unans	Cal	Ans	Unans	Cal	
Gemma4-E2B	Gemma4-E2B (base)	44.8	36.4	45.2	17.2	31.2	21.8	56.2	39.0	57.8	2.4	30.1	61.4	4.6	33.0
	+ GSPO (step 100)	47.8	42.4	43.5	29.3	36.4	26.4	64.2	45.3	62.7	4.9	33.8	67.5	13.8	40.7
	+ GSPO (step 200)	52.4	75.2	49.6	96.6	73.1	47.3	98.5	72.9	67.5	97.6	82.5	48.2	100.0	74.1
	+ GSPO (step 300)	61.6	79.0	57.4	94.0	75.7	67.3	98.5	82.9	68.7	92.7	80.7	53.0	98.5	75.7
	+ GSPO (step 400)	60.1	74.2	61.7	86.2	74.0	58.2	97.8	78.0	65.1	58.5	61.8	55.4	90.8	73.1
	+ GSPO (step 500)	56.8	57.4	54.8	39.7	47.2	49.1	91.2	70.2	60.2	22.0	41.1	66.3	43.1	54.7
	+ GSPO (step 600)	56.8	58.3	49.6	44.0	46.8	47.3	92.0	69.6	66.3	17.1	41.7	69.9	47.7	58.8
Gemma4-E4B	Gemma4-E4B (base)	63.7	52.6	65.2	37.9	51.6	41.8	67.9	54.9	74.7	4.9	39.8	79.5	15.4	47.5
	+ GSPO (step 100)	67.3	56.9	68.7	39.7	54.2	52.7	79.6	66.1	73.5	2.4	38.0	78.3	16.9	47.6
	+ GSPO (step 200)	68.8	57.7	67.8	37.9	52.9	54.5	80.3	67.4	75.9	2.4	39.2	81.9	18.5	50.2
	+ GSPO (step 300)	68.8	62.0	67.0	44.8	55.9	58.2	91.2	74.7	75.9	9.8	42.8	78.3	26.2	52.2
	+ GSPO (step 400)	68.3	60.7	69.6	40.5	55.0	55.5	92.7	74.1	74.7	9.8	42.2	77.1	20.0	48.6
	+ GSPO (step 500)	66.8	57.1	65.2	36.2	50.7	54.5	82.5	68.5	77.1	4.9	41.0	74.7	20.0	47.3
	+ GSPO (step 600)	67.3	59.4	67.8	37.9	52.9	52.7	89.8	71.3	74.7	4.9	39.8	78.3	24.6	51.5

Table S.4: Omni-Persona Benchmark Results: Gemma4 Series. (Note: **BS**: BERTScore, **Add. Metrics**: Additional Metrics, **1-FA**: 1-FalseAbs, **TA**: TrueAbs, **AA**: AbsAvg, **MLen**: Mean Length)

Model	Accuracy (Cal)			Generation Quality				Avg	Add. Metrics ($\times 100$)			MLen
	Overall	Ans	Unans	Abs-F1	ROUGE-L	Tok-F1	BS		1-FA	TA	Avg	
Gemma4-E2B (base)	36.8	44.8	28.1	40.2	54.4	9.3	80.0	41.9	89.0	28.1	58.6	85.8
+ GSPO (step 100)	42.7	47.8	37.1	46.8	51.3	8.6	80.0	44.9	80.6	37.1	58.8	80.1
+ GSPO (step 200)	74.3	52.4	98.1	64.8	6.6	3.2	80.3	54.2	4.1	98.1	51.1	17.7
+ GSPO (step 300)	78.3	61.6	96.4	64.6	4.2	2.6	80.6	55.5	6.4	96.4	51.4	10.8
+ GSPO (step 400)	73.6	60.1	88.3	63.5	11.3	3.4	80.5	54.4	17.7	88.3	53.0	21.2
+ GSPO (step 500)	57.3	56.8	57.9	57.1	38.1	6.8	80.1	50.6	58.6	57.9	58.3	61.2
+ GSPO (step 600)	58.3	56.8	59.9	57.6	37.4	6.5	80.2	51.0	56.0	59.9	58.0	57.8
Gemma4-E4B (base)	53.1	63.7	41.5	50.1	58.6	15.1	84.5	52.4	77.8	41.5	59.6	53.6
+ GSPO (step 100)	57.3	67.3	46.5	53.7	57.6	14.8	84.5	54.5	75.5	46.5	61.0	50.4
+ GSPO (step 200)	58.1	68.8	46.5	53.8	58.9	14.8	84.4	55.1	75.7	46.5	61.1	57.3
+ GSPO (step 300)	62.3	68.8	55.2	60.4	58.6	14.1	84.3	57.7	74.7	55.2	64.9	57.8
+ GSPO (step 400)	61.1	68.3	53.2	58.8	57.9	14.2	84.2	56.8	74.4	53.2	63.8	58.8
+ GSPO (step 500)	57.5	66.8	47.4	54.5	58.6	14.6	84.4	54.8	75.7	48.3	62.0	57.6
+ GSPO (step 600)	59.7	67.3	51.5	57.2	57.1	14.3	84.4	55.9	73.7	51.5	62.6	53.7

- Model scale enhances RL stability.** Larger backbones (e.g., 4.5B) handle RL updates more reliably, whereas smaller models are highly sensitive to specific reward settings. This suggests that the effectiveness of RLVR depends on the backbone size, and reward mechanisms should be carefully adjusted to match the model’s capacity.
- Textual reasoning remains a bottleneck for the Gemma4 family.** Models in this family exhibit a recurring performance dip on tasks requiring complex textual reasoning (specifically T2T), indicating that post-training alone cannot fully suppress intrinsic hallucinations in these scenarios.

C.2 Alignment of Evaluation Metrics

LLM-as-a-judge evaluations and traditional lexical/semantic metrics capture fundamentally distinct signals. Because their correspondence varies substantially across query types, this benchmark reports answerable QA quality and unanswerable abstention behavior separately.

Why LLM-as-a-judge is needed to evaluate free-form responses: Figure S.4 demonstrates that content-overlap metrics primarily explain performance on answerable queries, but fail to capture abstention behavior on unanswerable queries. That is, content quality and abstention capability are distinct evaluation axes. While answerable recall increases with ROUGE-L, indicating that lexical overlap partially reflects answer quality, unanswerable recall shows little association with it. This suggests that abstention behavior cannot be inferred from traditional content metrics and must be evaluated separately.

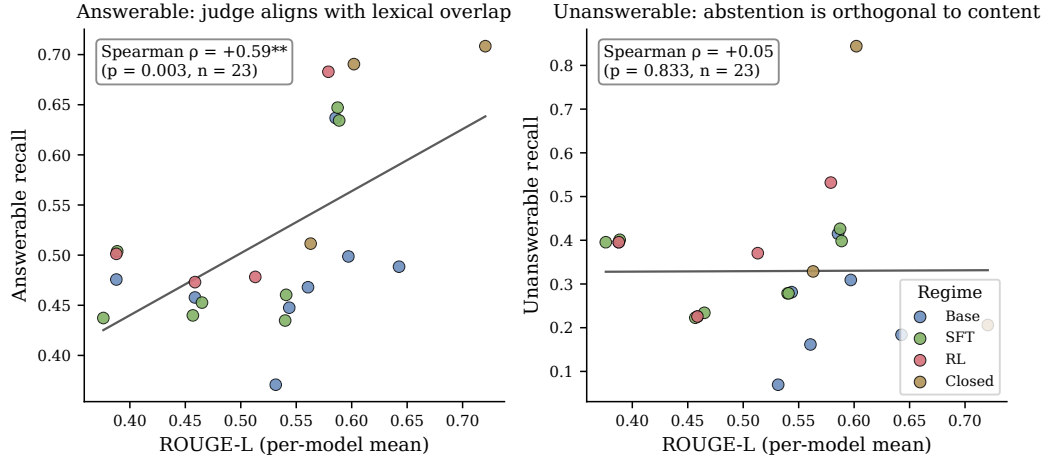


Figure S.4: Metric alignment across post-training regimes. The plot compares judge scores and lexical overlap for answerable and unanswerable queries. Crucially, it demonstrates the absence of content-induced bias in unanswerable cases, showing that abstention behavior is independent of content overlap.

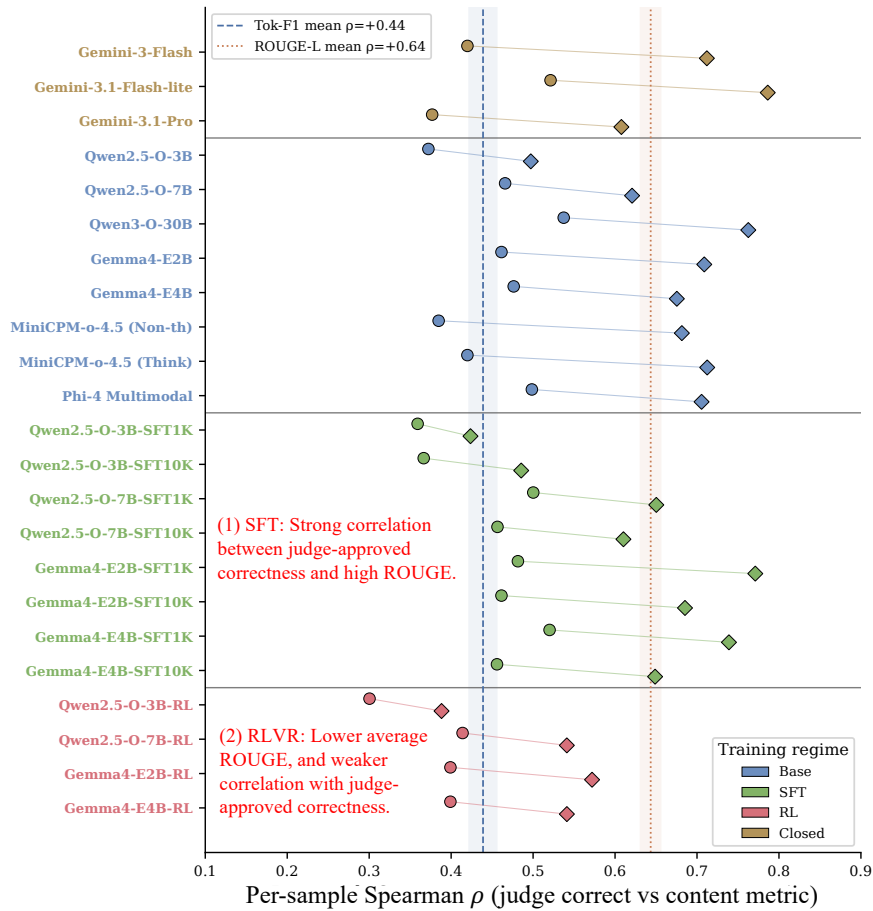


Figure S.5: Per-model agreement on answerable items only between judge and content metrics (circle = Tok-F1, diamond = ROUGE-L)

Table S.5: Summary of evaluation metrics used in the Omni-Persona benchmark. All scores are reported with a higher-is-better polarity; MeanLen is descriptive and is excluded from Avg.

Metric	Target Subset	Description
Overall	All	Total recall across both answerable and unanswerable scenarios.
Ans	Answerable	LLM-as-a-judge correctness rate.
Unans	Unanswerable	Successful abstention rate via keyword matching.
ROUGE-L	Answerable	Average LCS-based recall (sequence-sensitive lexical overlap).
Tok-F1	Answerable	Average token-level F1 (bag-of-words lexical overlap).
BERTScore	Answerable	Average BERTScore F1 (semantic embedding similarity).
Abs-F1	All	Harmonic mean of abstention precision and recall; penalizes over-abstention.
1 – FA	Answerable	Complement of FalseAbs: fraction of answerable queries the model does <i>not</i> answer with an abstention keyword.
TA	Unanswerable	TrueAbs: fraction of unanswerable queries on which the model correctly abstains.
Avg	Composite	Simple mean of the nine higher-is-better scores above (Overall, Ans, Unans, Abs-F1, ROUGE-L, Tok-F1, BERTScore, 1 – FA, TA).
MeanLen	All	Mean response length in whitespace-tokenized words (descriptive; not part of Avg).

Evaluation Metrics

Dual-Axis Recall

We measure basic task completion via a dual-axis accuracy protocol that decouples successful answering from appropriate abstention.

- **Ans:** Proportion of the 391 answerable queries judged correct by an LLM-as-a-judge.
- **Unans:** Proportion of the 359 unanswerable queries for which the model triggers an abstention keyword (e.g., “I can’t decide”).
- **Overall:** Combined accuracy across all 750 items.

Generation Quality (Answerable Only)

To assess the lexical and semantic fidelity of generated responses against the ground truth, we apply three established metrics on the answerable subset.

- **ROUGE-L Recall:** Structural overlap via the Longest Common Subsequence (LCS); strict on token order, penalizing valid paraphrasing.
- **Token F1:** Harmonic mean of token-level precision and recall under a bag-of-words representation, order-independent.
- **BERTScore F1:** Cosine similarity in the BERT embedding space, rewarding semantic equivalence beyond exact lexical matching.

Calibrated Abstention (Abs-F1). Unanswerable recall alone is gameable by always abstaining. **Abs-F1** closes this loophole by treating unanswerable queries as the positive class: TP are correct abstentions, FP are erroneous abstentions on answerable items, and Abs-F1 is the harmonic mean of the resulting precision and recall. A high Abs-F1 thus reflects calibrated confidence, rejecting unanswerable queries without sacrificing valid answers.

Anti-Hallucination Guardrails (1 – FA, TA). We report the two raw abstention rates jointly so that the *direction* of a calibration failure is visible: 1 – FA drops when the model *over-abstains* on answerable items (e.g., RL abstention-shortcut hacking), while TA drops when the model *hallucinates* a grounded answer on unanswerable items. A model that maximizes one through blanket behavior is therefore automatically penalized by the other.

Composite (Avg) and Length (MeanLen). **Avg** is the simple mean of the nine higher-is-better scores listed above; it flags single-axis policies (e.g., universal abstention) that boost one metric while collapsing others. **MeanLen** is the mean response length in whitespace-tokenized words and is reported descriptively only, since it has neither a $[0, 1]$ range nor a clear quality polarity.

Table S.6: Comprehensive Error Analysis across Answerable and Unanswerable Queries. This table illustrates the quality-personalization trade-off inherent in post-training.

Model	Answerable ($N = 391$)		Unanswerable ($N = 359$)	
	False Abst.	FA Rate ($\downarrow$)	Genuine Hall.	Hall. Rate ($\downarrow$)
Gemma4-E2B (Base)	43	11.0%	258	71.9%
Gemma4-E2B-SFT	45	11.5%	259	72.1%
Gemma4-E2B-RL	76	19.4%	226	63.0%
Gemma4-E4B (Base)	87	22.3%	210	58.5%
Gemma4-E4B-SFT	77	19.7%	216	60.2%
Gemma4-E4B-RL	100	25.6%	168	46.8%
Qwen2.5-Omni-3B	97	24.8%	217	60.4%
Gemini-3.1-Pro	112	28.6%	56	15.6%
<i>MiniCPM-o 4.5</i>	7	1.8%	334	93.0%

Key Finding 4: RLVR Trades Lexical Overlap for Targeted Personalization

Unlike SFT, which preserves the lexical fidelity of the base policy, RLVR optimization degrades agreement with traditional lexical metrics like ROUGE-L. This reveals a clear tradeoff: RL successfully improves personalization capabilities, but at the cost of conventional lexical overlap.

Per-Sample Metric Reliability: As illustrated in Figure S.5, ROUGE-L exhibits stronger agreement with LLM-as-a-judge evaluations than token-level F1 on answerable items across all models. While strong closed-source and base models maintain high concordance between the judge and lexical metrics, RL-tuned models display noticeably weaker alignment. This divergence indicates that reward optimization shifts the model’s outputs away from traditional lexical overlap. Conversely, SFT models largely preserve the judge-metric alignment of their base counterparts, suggesting that supervised fine-tuning does not induce significant lexical deviation from the original policy rather than RLVR.

C.3 Unveiling the Trade-offs: Calibrated Accuracy and Behavioral Shifts

Lexical Metrics Validate, but Cannot Replace, the LLM Judge. We first examine whether our LLM-as-a-judge verdicts are consistent with reference-based lexical signals. For each model, we partition answerable predictions ($N = 391$) by judge verdict and recompute lexical scores within each partition. Across all evaluated backbones, predictions judged as CORRECT consistently achieve substantially higher ROUGE-L than those judged as WRONG, typically by a factor of 2–4. This pattern suggests that the judge is aligned with the same reference-overlap signal captured by conventional lexical metrics. However, lexical overlap remains too sparse for free-form personalized QA. Even the highest overall ROUGE-L remains low, and threshold-based accuracy using $\text{ROUGE-L} \geq 0.5$ stays below 11% across models. Thus, lexical metrics provide a useful sanity check for judge reliability, but they cannot serve as the primary evaluation criterion. This motivates a calibrated, behavior-aware metric that accounts for both grounded answering and appropriate abstention.

Lexical Overlap and Calibrated Accuracy Disagree. A direct cross-model comparison further shows that ROUGE-L alone can misrepresent personalization quality. For example, Gemini-3.1-Pro achieves the strongest judge-based accuracy and unanswerable abstention accuracy, yet its ROUGE-L is lower than that of weaker models such as Gemini-3-Flash. Conversely, models with poor abstention behavior may receive low ROUGE-L, but the underlying failure mode is often over-answering rather than poor linguistic overlap. This shows that lexical overlap does not capture the deployment-relevant behavior required for personalization: answering when evidence is available and abstaining when it is not. Calibrated Accuracy addresses this gap by jointly measuring answerable recall and unanswerable abstention accuracy.

Post-training Regimes Show Different Behavioral Fingerprints. From this calibrated perspective, SFT and RLVR produce qualitatively different behavioral shifts (Table S.6). SFT-10K largely preserves the base-model behavior: ROUGE-L changes only marginally, and the Genuine Hallucination

Rate remains nearly unchanged. This suggests that broader supervised data alone does not reliably reshape the model’s abstention boundary in open-ended personalization settings. In contrast, RLVR more directly reshapes this boundary. It reduces genuine hallucinations and improves abstention accuracy, but it also increases false abstention on answerable queries and slightly lowers ROUGE-L. This reflects a clear trade-off under our current reward design: outcome-level rewards improve calibration, but may also encourage overly conservative abstention. The closed-source models illustrate the same tension from another angle. Gemini-3.1-Pro achieves strong calibrated behavior, whereas MiniCPM-o-4.5 tends toward over-answering. These extremes confirm that single-axis metrics are insufficient: lexical quality, recall, hallucination, and abstention behavior can move independently. Calibrated Accuracy is therefore an appropriate headline metric for personalized omnimodal QA because it jointly penalizes blanket abstention and unsupported guessing.

D Additional Experimental Configurations

D.1 Details on SFT Implementations

SFT Training Objective. We train the model on $\mathcal{D}_{\text{SFT}}$, a dataset of triplets $(q, \mathcal{C}, y^*)$, where q is a query, $\mathcal{C}$ is omnimodal contexts, and y^* is the ground-truth response. We minimize the standard autoregressive negative log-likelihood:

$$\mathcal{L}_{\text{SFT}}(\theta) = -\mathbb{E}_{(q, \mathcal{C}, y^*) \sim \mathcal{D}_{\text{SFT}}} \sum_{t=1}^{|y^*|} \log \pi_{\theta}(y_t^* \mid q, \mathcal{C}, y_{<t}^*). \quad (\text{S.1})$$

SFT Configurations. We fine-tune four backbones via ms-swift: Qwen2.5-Omni (3B/7B) and Gemma4 (E2B/E4B). All SFT runs use $\text{lr} = 2 \times 10^{-5}$ over 3 epochs with vision/audio encoders frozen. We apply LoRA ($r = 64, \alpha = 128$) for the Qwen backbones and a more compact LoRA ($r = 32, \alpha = 64$) for the Gemma4 models.

SFT Data Corpus Construction. The 10K-scale SFT corpus (detailed in Table S.8) encompasses 12 distinct task types explicitly designed to enforce robust omnimodal alignment. The dataset comprehensively integrates foundational grounding tasks, audio-centric scenarios (*e.g.*, conversational and event sounds), and missing-modality variants that train the model to recognize and verbalize when required context is absent. To mitigate positional and cross-modal shortcut biases, we apply rigorous augmentations such as context reordering, distractor replacement, and modality swapping. Contextual memory is sourced from CoViP dialogues [9] and CUPID biographies [34], with ground-truth visual captions rigorously filtered by an MCQA accuracy threshold ($\geq 50\%$). A downscaled 1K subset is also provided for experimental efficiency (Table S.7).

Training Data Corpus Overview. Our experimental pipeline comprises three distinct data stages (see Table S.10). The SFT corpus (1K and 10K variants) provides broad modality-alignment supervision, examining the impact of data scaling. Subsequently, the RLVR corpus supplies VR signals focused on two core grounding tasks: audio and image perception, and text-grounded retrieval. Notably, across both the SFT and RLVR regimes, we ensure that approximately 20% of the training mixtures consist of absent-persona (no-GT) samples. This deliberate inclusion enforces the model’s ability to demonstrate calibrated abstention when confronted with unanswerable queries. The dataset distribution across each training stage is illustrated in Figure S.6.

D.2 Details on RLVR Implementations

We adopt Group Sequence Policy Optimization (GSPO; 37) as the base optimizer for our RLVR, with a moderate KL regularization coefficient $\beta = 0.04$.

RLVR Training Objective. We optimize π_{θ} to maximize a verifiable reward while penalizing the KL divergence [38, 37] from the reference policy π_{ref} :

$$\max_{\theta} \mathbb{E}_{(q, \mathcal{C}) \sim \mathcal{D}_{\text{tr}}} \mathbb{E}_{y \sim \pi_{\theta}(\cdot \mid q, \mathcal{C})} [r(y, q, \mathcal{C}) - \beta \text{D}_{\text{KL}}(\pi_{\theta}(\cdot \mid q, \mathcal{C}) \parallel \pi_{\text{ref}}(\cdot \mid q, \mathcal{C}))] \quad (\text{S.2})$$

here, q is a query, $\mathcal{C}$ is omnimodal contexts, y is the model response. For each training instance, the binary reward $r(y, q, \mathcal{C}) \in \{0, 1\}$ is specifically determined based on the success of y in the assigned task as follows:

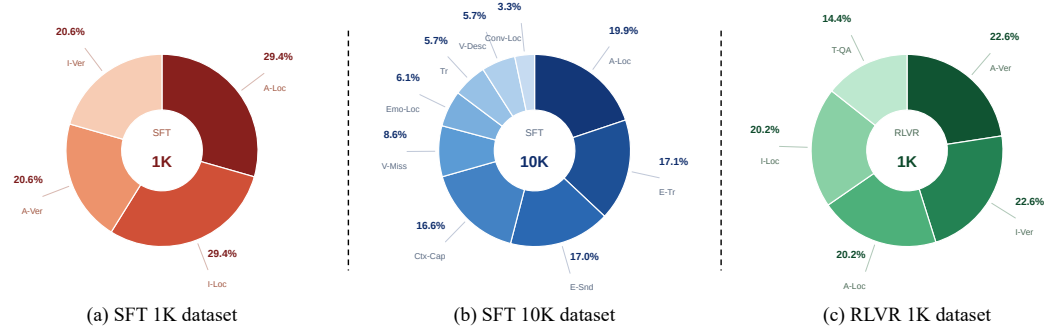


Figure S.6: **Post-training data strategy.** Visualization of the task coverage and data distributions across the SFT (1K), SFT (10K), and RLVR training regimes.

Table S.7: **SFT (1K) task distribution.** Overview of the task proportions and corresponding abbreviations utilized in our post-training ablations.

Task name	Sample query text
Audio localization	Select which reference audio panel contains the same speaker as the query audio.
Image localization	Select which reference image panel contains the same person/object as the query image.
Audio verification	Answer yes/no on whether the query audio matches the referenced speaker/audio identity.
Image verification	Answer yes/no on whether the query image matches the referenced person/object identity.

Table S.8: **SFT (10K) task distribution.** Overview of the task proportions and corresponding abbreviations utilized in our post-training ablations.

Task name	Sample query text
Audio localization	Select the reference speaker/audio panel that matches the query audio.
Event-aware transcription	Transcribe speech while handling an added environmental sound event.
Event sound MCQA	Identify the environmental sound mixed with speech from multiple choices.
Contextualized captioning	Describe the query image using the matching memory/context panel when relevant.
Missing-image visual description	Describe visual information when the target image modality is absent.
Emotion speaker localization	Find which reference speaker matches the query speech/emotion cue.
Speech transcription	Convert the target speech audio into text.
Visual description	Generate a grounded description of the target image.
Conversation speaker localization	Identify the matching speaker/person from dialogue-conditioned references.

Table S.9: **Task distribution in the RLVR (1K) dataset.** This table summarizes the task abbreviations used in the RLVR training distribution. For all referential tasks, the target responses are restricted to binary "Yes" or "No" labels.

Task name	Sample query text
Audio verification	Decide whether the query audio matches the referenced speaker/audio identity.
Image verification	Decide whether the query image matches the referenced person/object identity.
Audio localization	Select which reference audio panel contains the same speaker as the query audio.
Image localization	Select which reference image panel contains the same person/object as the query image.
Text QA	Answer a context-grounded question or abstain when the answer is not supported.

Table S.10: Overview of data configurations for training and evaluation phases.

Stage	Size	# of Contexts	Tasks	Source Modality		
				Image	Text	Audio
SFT Corpus	1K / 10K	3	12 foundational task types	Synthetic-only	Synthetic (Phases A-C, GPT-5.4)	Real & Syn. (80%)
RLVR Corpus	1K	3	3 core grounding tasks (Visual ID, Voice ID, Text QA)	Synthetic-only	Synthetic (GPT-5.4)	Real & Syn. (80%)
Eval Benchmark	750	4	18 fine-grained tasks (4 groups)	Real-only	Synthetic (GPT-5.4)	Real & Syn. (20%)

RLVR Configurations. Following SFT, we optimize the RLVR objective using GSPO [37] on 1K samples. Key hyperparameters include: $G = 4$ generations, $\beta = 0.04$, $\text{lr} = 1 \times 10^{-5}$, and LoRA $r = 32, \alpha = 64$. Similar to SFT, we freeze all perceptual encoders to maintain pre-trained representations. **Verifiable Reward Designs.** Our reward function targets two core capabilities: *perception* and *retrieval*. Each training instance belongs to one of these two task types, and the corresponding reward is applied:

$$r_{\text{base}} = \begin{cases} r_{\text{perc}}(y, q, \mathcal{C}), & \text{if } t(q) = \text{PERC}, \\ r_{\text{retr}}(y, q, \mathcal{C}), & \text{if } t(q) = \text{RETR}. \end{cases} \quad (\text{S.3})$$

Both rewards are binary, assigning 1 for a successful response and 0 otherwise.

Perception VR. Perception VR trains the model to match a target perceptual signal, such as an image or an audio clip. Each query provides only the target modality, preventing the model from relying on shortcut cues from other modalities, such as names in nearby text. The reward is computed by comparing the model’s binary answer, such as yes or no, with the ground-truth label. This provides a simple deterministic signal without requiring an LLM judge.

Retrieval VR. Retrieval VR trains the model to produce grounded answers in open-ended QA settings. If the target persona is present in the context $\mathcal{C}$, an LLM judge (GPT-5.4-mini) checks whether the model retrieves the correct information. If the target persona is absent, we reward appropriate abstention using the same lexical abstention rule used in evaluation, with the judge called only when needed. By including absent-persona cases during training, the model learns not only what to retrieve, but also when to abstain.

Safeguards Against Reward Hacking. We include several safeguards to reduce reward hacking during RL fine-tuning. Outputs with clear degeneration patterns, such as repeated 4-grams, character-level spam, or sentence loops, receive zero reward. We also parse LLM-judge outputs with strict exact matching, so that only the verdict correct is rewarded and strings such as incorrect cannot be mistakenly counted as positive. Invalid ground-truth labels are treated as explicit errors rather than ignored silently. These safeguards make the reward signal more reliable while keeping the training procedure simple.

On-Policy RLVR algorithms. Given a prompt x , group size G , and sampled responses $\{y_i\}_{i=1}^G$ with group-normalized advantages $\hat{A}_i$, the two objectives differ in the granularity at which the importance ratio is computed and clipped:

$$\mathcal{J}_{\text{GRPO}}(\theta) = \mathbb{E} \left[\frac{1}{G} \sum_{i=1}^G \frac{1}{|y_i|} \sum_{t=1}^{|y_i|} \min \left(r_{i,t}(\theta) \hat{A}_i, \text{clip}(r_{i,t}(\theta), 1 \pm \varepsilon) \hat{A}_i \right) \right] - \beta \mathbb{D}_{\text{KL}}[\pi_\theta \parallel \pi_{\text{ref}}], \quad (\text{S.4})$$

$$\mathcal{J}_{\text{GSPO}}(\theta) = \mathbb{E} \left[\frac{1}{G} \sum_{i=1}^G \min \left(s_i(\theta) \hat{A}_i, \text{clip}(s_i(\theta), 1 \pm \varepsilon) \hat{A}_i \right) \right], \quad (\text{S.5})$$

where $r_{i,t}(\theta) = \pi_\theta(y_{i,t} \mid x, y_{i,<t}) / \pi_{\text{old}}(y_{i,t} \mid x, y_{i,<t})$ is the per-token ratio, and

$$s_i(\theta) = \left(\frac{\pi_\theta(y_i \mid x)}{\pi_{\text{old}}(y_i \mid x)} \right)^{1/|y_i|} \quad (\text{S.6})$$

is the length-normalized sequence-level ratio. Both objectives share the same KL reference π_{ref} ; the substantive distinction lies in Eq. (S.6) and in whether clipping is applied per-token (Eq. (S.4)) or per-sequence (Eq. (S.5)).

Why Sequence-Level Optimization Suits our RLVR. Our RLVR verifier combines rule-based binary verification (e.g., ‘yes’/‘no’ compliance) with an LLM-as-judge component, producing a reward signal that is defined at the sequence outcome level rather than at the token level. Under GRPO [38], a single noisy high-reward rollout propagates its advantage through per-token ratios; consequently, tokens that happen to diverge from π_{old} receive disproportionately large updates, even if they are not causally responsible for the binary outcome. In contrast, GSPO [37] aggregates the ratio over the full response before clipping, ensuring that per-token fluctuations cancel out and only the geometric-mean deviation of the sequence survives. This alignment between clipping and reward granularity significantly reduces gradient variance in the presence of judge-model false positives and rule-based shortcut exploitation.

Table S.11: Keyword and phrase list used for lexical abstention detection.

Abstention Keywords / Target Surface Forms	
cannot determine	cannot be determined
not enough information	insufficient information
cannot answer	unable to determine
don't know from	do not know from
the provided context does not	not provided in the context
no information in the context	context does not contain
cannot identify	i cannot tell

Role of $\beta = 0.04$ in GSPO. We use $\beta=0.04$ as a conservative choice given the LLM-as-a-judge reward noise, noting that GSPO’s original formulation omits KL regularization. RLVR begins from a LoRA-adapted policy without any SFT warmup on the target persona distribution. At step 0, $\pi_\theta \approx \pi_{\text{ref}}$ and $\mathbb{D}_{\text{KL}} \approx 0$, so the choice of β governs how much cumulative drift is tolerated once advantages become informative. We found that $\beta = 0$ (unconstrained) permits rapid drift, which in preliminary runs led to format collapse and over-assertive answering on unanswerable prompts. In contrast, $\beta = 0.04$ preserved the group-relative advantage signal while bounding drift to a trustful regime.

E Limitations and Broader Impacts

Synthetic-Real Domain Bias in Post-Training. Real-world face images and voice recordings of consistent identities are difficult to obtain at the scale required for systematic SFT and RLVR training, particularly given privacy, consent, and licensing constraints. We therefore train on synthetic personas (TTS-generated voice clips and generated facial images) while reserving real images strictly for the Omni-Persona evaluation benchmark. This deliberate split preserves benchmark realism and also provides a natural testbed for studying how well post-training generalizes from synthetic personas to real-world distributions, a question we view as important for future omnimodal personalization research.

Difficulty of Evaluation-Aligned In-Domain SFT Data. One could alternatively use an SFT dataset that closely matches the evaluation distribution. However, constructing such a dataset at scale is a significant challenge. Generating high-quality answers without accidentally exposing test-set information is difficult, and using test-style queries during training risks benchmark contamination. Therefore, we treat our SFT results as an analysis of broad-coverage training. Developing a reliable method to create high-quality, in-domain data while maintaining benchmark integrity remains an important goal for future research.

Scope of Reward Design Ablations. Our RLVR pipeline includes several stability filters, such as 4-gram repetition, character diversity, and sentence-repetition checks, which effectively prevent severe degeneration patterns that can occur during RL training on thinking-style models. Fine-grained reward shaping for balancing grounding and abstention is a distinct and complementary direction; we leave such ablations to future work. Importantly, we view the over-conservatism surfaced by our FA metric as a diagnostic strength of Omni-Persona, rather than as evidence that RLVR is inherently prone to this behavior, and we expect it to be addressable with more targeted reward design.

Limitations of Lexical Abstention Detection To determine whether a model abstains on unanswerable queries, we use a simple lexical matching rule. A response is classified as an abstention if it contains any predefined phrase listed in Table S.11. The same rule is applied both during RLVR training, where it defines the abstention-related reward signal, and at evaluation time, making abstention evaluation directly comparable without requiring an additional judge model. This lexical design may undercount valid abstentions, particularly when base or SFT models use phrases outside the predefined keyword list. Although the evaluation prompt in Table S.16 instructs models to use “I cannot determine that from the provided context” for unsupported answers, models may still produce paraphrases or partial uncertainty statements that are not captured by the detector.

Privacy and Consent. Omni-Persona is constructed from publicly available or synthetically generated user signals, including face images, voice samples, and biographical text, and does not include personally identifiable information from real individuals without consent. However, systems built for real-user personalization must obtain explicit consent for collecting, storing, and processing biometric signals such as voice and facial imagery.

Bias and Representation. Synthetic persona generation may underrepresent certain demographic groups, leading to uneven personalization quality across users. While our benchmark includes hard distractors and retrieval noise to test robustness, fairness across multilingual voice, facial, and linguistic attributes remains an important direction for future evaluation.

Future Directions. A natural next step is to extend Omni-Persona to on-device personalization benchmarks built from unstructured omnimodal user signals, such as daily voice memos, image galleries, and textual interaction histories. Future benchmarks should also evaluate agentic tool-use, where models execute complex cross-modal actions grounded in locally stored user context.

F Details on Evaluation Dataset and Metrics

F.1 Task Overview

Omni-Persona instantiates the four PMG scenarios into 18 fine-grained sub-tasks organized into four distinct groups, each covering a different combination of query and context modalities. The groups are structured as follows:

- *Group 1 (I2I; 5 tasks):* A face-image query is matched against the persona’s stored face images to ground visual identity, and the model then retrieves one of five persona attributes: *Biography* (text), *Dialogue* (voice), *Appearance* (image), *Emotion* (voice), or *Environment* (voice). All sub-tasks except *Appearance* require a visual-to-text or visual-to-audio bridge.
- *Group 2 (A2A; 5 tasks):* A voice query is matched against stored speaker samples to identify the persona, then sweeps the same five retrieval targets as Group 1. Sub-task 2-c (*Appearance*) further requires an audio-to-visual bridge, a capability absent in purely unimodal retrieval.
- *Group 3 (T2T; 4 tasks):* Identity is resolved through semantic matching between a textual query and the persona’s stored textual profile, covering four retrieval targets: *Biography* (text), *Appearance* (image), *Emotion* (text), and *Environment* (text). Sub-task 3-b additionally crosses modality by retrieving a visual appearance description from a purely textual match. The *Dialogue* target is intentionally omitted, as text-to-text conversational matching risks collapsing into shallow keyword overlap.
- *Group 4 (T2Any; 4 tasks):* Forms the benchmark’s most demanding regime: a textual description of conversational content must be matched to the persona whose stored conversational audio semantically corresponds to it, without any explicit speaker cues. Retrieval targets span *Biography* (text), *Appearance* (image), *Emotion* (voice), and *Environment* (text). Sub-task 4-b requires a three-hop cross-modal path from the text query through audio-based identity matching to visual appearance retrieval, analogous to 2-c.

Across all four groups, every sub-task is paired with a no-GT (absent-persona) variant that requires the model to perform structured abstention, as summarized in Table S.15.

F.2 Benchmark Complexity and Design Principles

Our proposed benchmark is systematically more challenging in several respects. The evaluation distractors are deliberately curated to be highly similar to the target, sharing specific vocal characteristics or visual resemblances, thereby requiring fine-grained cross-modal discrimination. Moreover, each evaluation context contains four interleaved image-audio-text entries, exposing models to dense and heterogeneous multimodal signals. Detailed per-scenario prompt templates and query-construction procedures are provided in Appendix H. Crucially, we introduce a deliberate distribution shift between training and evaluation to assess true generalization: training uses three context entries paired with



Figure S.7: **Qualitative image examples.** Sample images utilized during the training and evaluation phases. Note that synthetic images are strictly employed for SFT and RL training, whereas real images are reserved exclusively for the Omni-Persona benchmark.

synthesized CoViP images, whereas evaluation uses four context entries paired exclusively with real images.

E.3 Evaluation Dataset Configurations

Construction of Image–Text Pairs. To formulate diagnostic tasks for omnimodal personalization, it is essential to have multiple images of the same individual to facilitate cross-context identity reasoning. To this end, we sample person-related contexts from the evaluation split of CoViP, ensuring each individual appears as the query subject across diverse scenarios. This process yields 250 context–query pairs, each comprising a query image and four interleaved context entries, where each entry consists of an identity image paired with descriptive text.

To systematically construct the textual elements of the Omni-Persona benchmark, we refine the pre-generated dialogues through a multi-stage pipeline. First, a model extracts explicit personal attributes from the raw dialogue. It then plausibly imputes any missing traits and restructures the extracted information into concise personal profiles (*e.g.*, in biography or dialogue format), strictly limited to 1–2 sentences per concept. By bifurcating scenarios into answerable and unanswerable (no-GT) cases, we synthesize targeted query prompts accordingly: for answerable cases, a GT answer is generated based on the target context; for unanswerable cases, the target is formatted to elicit structured abstention. All synthesis procedures are conducted using the proprietary frontier model GPT-5.4.

Audio Modality Construction. The audio modality consists of a balanced mixture of synthetic voice samples and real-world recordings, covering a total of 450 distinct speakers.

Synthetic Data. For synthetic samples, we use chatterbox[§] to generate high-fidelity audio (24-bit, 24 kHz). For each speaker, we construct pairs of distinct clips that share the same voice identity while varying emotion, conversational setting, and metadata such as age and accent.

Real-world Data. Real-world conversational audio is curated from diverse corpora, including VoxMM [39], MELD [40], JL-corpus [41], and RAVDESS [42]. We extract 4–15 second segments and ensure expressive variation across two roles:

- **Reference Audio:** Uses relatively neutral or mild emotions, such as *neutral*, *calm*, and *happy*.
- **Emotional Utterance:** Uses more expressive and distinct emotions, such as *angry*, *sad*, *fearful*, *disgust*, and *surprised*.

Context–Query Pair Construction. To construct challenging audio distractors for visually grounded pairs, we use wav2vec2 [43] for automated gender detection. Each target voice is then paired with a *gender-aligned distractor* voice from either the synthetic or real-world subset. This setup prevents the model from relying on coarse gender cues and instead requires more fine-grained recognition of speaker-specific vocal characteristics.

[§]<https://github.com/resemble-ai/chatterbox>

Contexts			Brandon Kim is the young scriptwriter on the sci-fi series First Light in Los Angeles, always talking about believable alien languages and how good world-building makes a story click. He has that energetic ENFP, writers-room kind of presence.
			Willie is the bookstore assistant I ran into near the old bookstore by the river in Montoyamouth, the one who laughed about missing her own dropped scarf because she was lost in a poem. She seemed warm, thoughtful, and quietly absorbed.
			Tony was the bookstore assistant I remember from Kaylacheater market square, drifting between secondhand stalls while hunting for a rare Korean poetry collection. He came across as an INFP type, slender and a little absent-minded.
			Wei Chen is an entry-level biophysicist in Boston who studies protein interaction mechanisms and is always encouraging collaborative problem-solving in the lab. She also mentors undergrads and has a polished, approachable presence.
Query			Visual and audio cues are provided in the query
			This speaker in the query audio sounds familiar—what mood were they in? GT Ans: Warm, thoughtful, and quietly absorbed
Responses	Gemma4-E4B		
	The query image matches the appearance of Person 1 (Willie). The query audio is not provided, so I will rely on the visual match and the text description associated with Person 1. Willie, the bookstore assistant, seemed warm, thoughtful, and quietly absorbed when the speaker was lost in a poem. Auditory Signal Inferiority under Visual Dominance!		
	Gemma4-E4B + RLVR		
	The speaker in the query audio sounds warm and thoughtful, like Willie, the bookstore assistant I ran into near the old bookstore by the river in Montoyamouth. She seemed quietly absorbed in her own world, laughing about missing her own dropped scarf because she was lost in a poem.		

Figure S.8: **Qualitative comparison of model predictions given the query and its GT answer.** When the query includes both image and audio modalities and audio is the primary perceptual cue, Gemma4-E4B tends to overlook the audio signal and rely heavily on visual cues. In contrast, the RLVR-trained model grounds its response in the relevant audio signal.

In summary, the final Omni-Persona evaluation benchmark comprises approximately 750 items spanning 18 tasks across 4 scenario groups. This benchmark serves strictly as a held-out test set.

G Additional Results

Results on Other Evaluation Metrics. Complementary lexical and semantic metrics indicate that SFT preserves the base model’s generation quality, whereas RLVR trades a modest amount of lexical overlap (ROUGE-L, Token-F1) for the Cal gains reported above.

Other Personalization Benchmarks. We further evaluate on the CoViP downstream benchmark to situate our models within the broader visual personalization landscape. The results indicate that omnimodal and multimodal LLMs still lag behind dedicated VLMs in interleaved image-text processing, suggesting that vision-language understanding remains a bottleneck for general-purpose multimodal architectures.

Our RLVR is not optimized for this benchmark. Instead, its verifiable rewards target the core competencies emphasized in Omni-Persona: image-based identity matching, audio-based identity matching, and text-grounded QA. Since CoViP primarily evaluates captioning quality, gains on our objective do not necessarily translate directly to CoViP scores.

RLVR Ablation studies. Additionally, we present ablation studies to identify the optimal VR composition, hyperparameters, and on-policy algorithms, in Tables S.13 and S.14.

Table S.12: Answerable-only recall score performances on CoViP [9] evaluation tasks.

Models	LSD-F1		LAR		ITR	
	Direct	w/ CAG	Direct	w/ CAG	Direct	w/ CAG
Vision-Language Models						
Qwen3-VL-8B	29.8	48.8	17.4	19.6	9.40	6.80
Qwen3-VL-30B-A3B	25.6	42.1	7.60	16.8	8.80	0.40
Qwen3-VL-8B + CoViP	37.2	58.2	34.8	49.2	28.0	42.8
Omni & Multimodal LLMs						
Qwen2.5-Omni-3B	11.59	5.19	1.40	1.20	3.20	0.80
Qwen2.5-Omni-7B	8.52	9.88	1.60	4.00	2.80	2.60
Gemma4-E2B	0.99	1.97	0.80	1.40	3.80	1.00
Gemma4-E4B	10.67	41.36	0.40	0.40	2.00	0.00
+ RLVR (Ours)	21.25	43.00	0.60	0.40	1.80	0.00

Table S.13: Omni-Persona benchmark ablation results.

Model	Overall		I2I			A2A			T2T			T2Any		
	Ans	Cal	Ans	Unans	Cal	Ans	Unans	Cal	Ans	Unans	Cal	Ans	Unans	Cal
Gemma4-E2B (base)	44.8	36.4	45.2	17.2	31.2	21.8	56.2	39.0	57.8	2.4	30.1	61.4	4.6	33.0
+ Optimal Step	67.3	56.9	68.7	39.7	54.2	52.7	79.6	66.1	73.5	2.4	38.0	78.3	16.9	47.6
+ Textual-retrieval only	64.5	73.5	59.1	74.1	66.6	69.1	97.8	83.5	72.3	53.7	63.0	57.8	83.1	70.5
+ Perception matching only	45.8	36.7	45.2	11.2	28.2	21.8	57.7	39.7	59.0	2.4	30.7	65.1	9.2	37.1
+ Another algorithm (GRPO)	58.6	77.9	53.9	96.6	75.2	58.2	97.1	77.6	71.1	95.1	83.1	53.0	100.0	76.5
+ GSPO with $\beta = 0$	49.6	73.8	53.9	96.6	75.2	30.9	98.5	64.6	67.5	97.6	82.5	50.6	100.0	75.3

Table S.14: Omni-Persona Benchmark Results: Gemma4 Series. (Note: BS: BERTScore, Add. Metrics: Additional Metrics, 1-FA: 1-FalseAbs, TA: TrueAbs, AA: AbsAvg, MLen: Mean Length)

Model	Accuracy (Cal)			Generation Quality				Avg	Add. Metrics			MLen
	Overall	Ans	Unans	Abs-F1	ROUGE-L	Tok-F1	BS		1-FA	TA	Avg	
Gemma4-E2B (base)	36.8	44.8	28.1	40.2	54.4	9.3	80.0	41.9	89.0	28.1	58.6	85.8
+ Optimal step	42.4	47.8	37.0	46.8	51.3	8.6	80.0	44.9	80.6	37.0	58.8	80.1
+ Textual-retrieval only	73.1	64.5	82.5	63.3	19.4	4.5	80.9	55.4	28.1	82.5	55.3	25.6
+ Perception matching only	37.1	45.8	27.6	39.0	55.3	9.8	80.1	42.1	87.2	27.6	57.4	83.9
+ Another algorithm (GRPO)	77.1	58.6	97.2	64.8	8.6	3.5	80.6	55.8	5.4	97.2	51.3	14.9
+ GSPO with $\beta = 0$	72.8	49.6	98.1	64.6	15.7	5.0	81.6	55.3	3.1	98.1	50.6	22.8

H Detailed Task Taxonomy and Granularities

Table S.15: Fine-grained task taxonomy of the Omni-Persona benchmark: 4 matching scenario groups and 18 sub-tasks derived from the PMG formulation. Each group represents a primary matching scenario defined by its query and context modalities; sub-tasks within a group sweep the retrieval target modality. Symbols: I (image), T (text), A^v (audio/voice sample). Every sub-task additionally has an absent-persona (no-GT, $e_{q \rightarrow j}=0$, target=None) variant that requires structured abstention.

ID	Query	Context	Target	Task	Sample question
<i>Group 1 (I2I): Visual Identification (Query: I, Context: I)</i>					
1-a	I	I	T	Biography	Who is this person, and what is their job?
1-b	I	I	A^v	Dialogue recall	Looking at this person in the image, I'm trying to remember what they said before.
1-c	I	I	I	Appearance	This person in the image seems familiar. What did they look like?
1-d	I	I	A^v	Emotion	When I met this person in the image, what mood were they in?
1-e	I	I	A^v	Environment	Who is this person in the image, and where were we?
<i>Group 2 (A2A): Voice Identification (Query: A^v, Context: A^v)</i>					
2-a	A^v	A^v	T	Biography	Who is the speaker in the query audio, and what is their affiliation?
2-b	A^v	A^v	A^v	Dialogue recall	Whose voice is in the query audio, and what had they said earlier?
2-c	A^v	A^v	I	Appearance	I recognize this speaker from the audio. Can you describe their appearance?
2-d	A^v	A^v	A^v	Emotion	This speaker in the query audio sounds familiar. What mood were they in?
2-e	A^v	A^v	A^v	Environment	This speaker in the query audio belongs to someone I know. What was happening around them?
<i>Group 3 (T2T): Same-Modal Semantic (Query: T, Context: T)</i>					
3-a	T	T	T	Biography	The professional esports player, what was their occupation?
3-b	T	T	I	Appearance	I'm thinking of the bookstore café barista. What were they wearing?
3-c	T	T	T	Emotion	The freelance illustrator, how were they feeling?
3-d	T	T	T	Environment	The gallery assistant, where were we when that happened?
<i>Group 4 (T2Any): Cross-Modal Semantic (Query: T, Context: A^v)</i>					
4-a	T	T	T	Biography	The one who said that, what was their affiliation?
4-b	T	T	I	Appearance	The person who talked about a peaceful reunion over poetry and a dropped scarf, what did they look like?
4-c	T	T	A^v	Emotion	The person who talked about an unexpected rainy-day encounter outside an art gallery, how were they feeling then?
4-d	T	T	T	Environment	Can you identify the person who talked about a memorable fan encounter at a gaming convention, and tell me where we were?

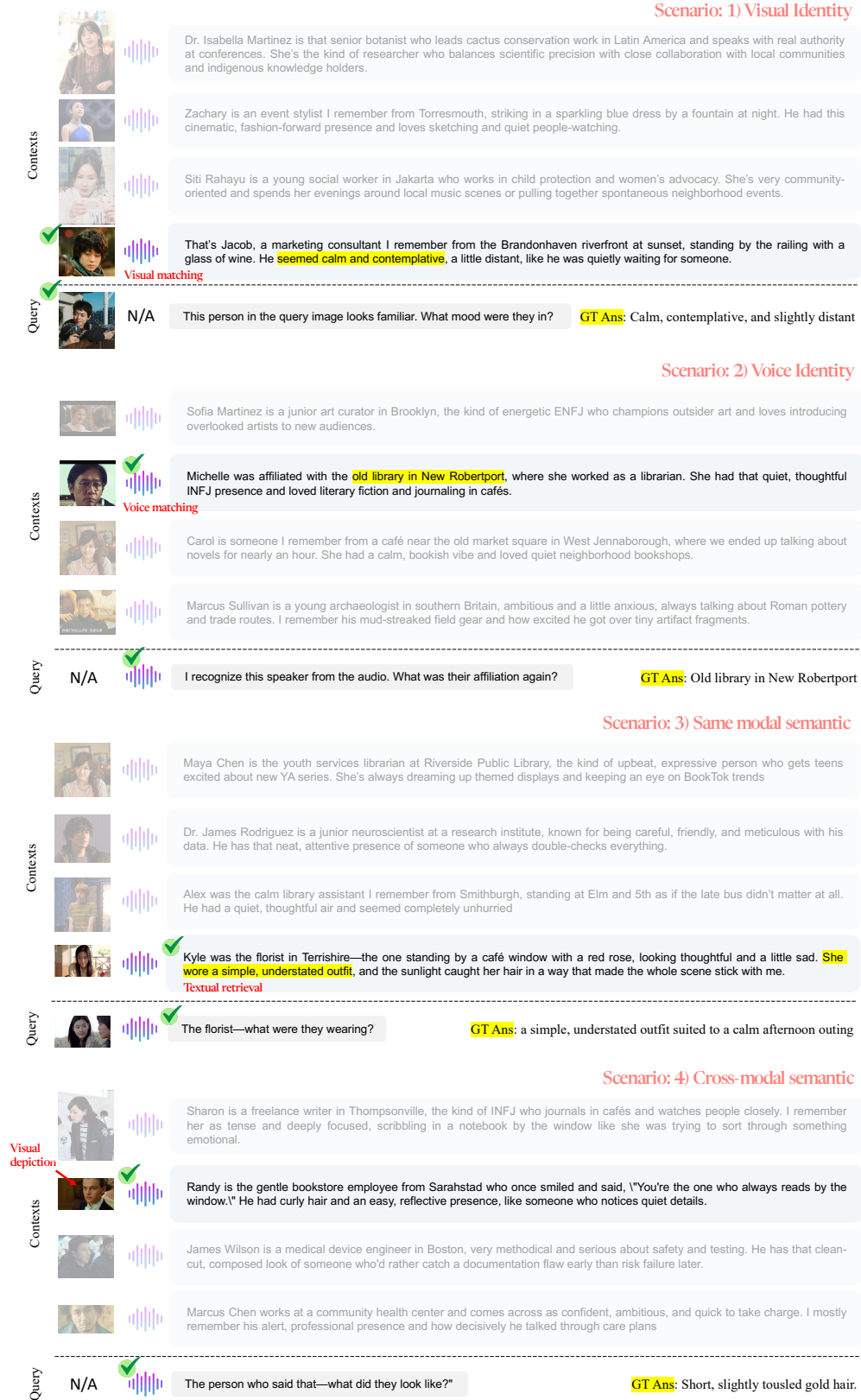


Figure S.9: Illustrative examples for each scenario within the Omni-Persona benchmark. Specifically, only answerable instances (where the target persona is present) are visualized in this figure.

I Used Templates for Dataset Construction

Table S.16: Fixed prompt template used for evaluation in our Omni-Persona benchmark.

User Prompt Prompt:

You are a personal memory assistant for a multimodal personalization benchmark.

[Context Setup]

You will receive memory contexts about four different people (Context 0 through Context 3). Each context may contain some combination of:

- A photo of the person (image)
- A voice recording or spoken clip of the person (audio)
- A text description with personal details such as name, job, hobbies, personality, location, etc.

After the four memory contexts, you will receive a Query section that may include:

- A query image showing a person (use it to identify which context person this is)
- A query audio clip of a person speaking (use it to identify which context person this is)
- A question asking for specific information about that person

[Task]

1. Carefully examine all four contexts, attending to every available modality.
2. Use the query image, audio, or text to identify which context (Person 0-3) the query refers to.
 - If the query includes an image, match the face or appearance to the context images.
 - If the query includes an audio clip, match the voice to the context audio recordings.
 - If the query is text-only, use the semantic description to identify the correct person.
3. Answer the question using only the information from the matched context.

[Response Rules]

1. Before answering, briefly reason through which person matches the query and what relevant information their context contains.
2. Provide a specific and informative answer in **1-3 sentences**, including all relevant details available in the matched context.
3. Base the answer solely on the provided contexts; do not hallucinate or infer beyond what is explicitly given.
4. Do not repeat the question.
5. Use **I cannot determine that from the provided context.** only as a last resort, when the requested information is genuinely absent from all modalities of the matched person's context.
6. If any modality provides a partial or indirect answer, use it rather than abstaining.

[Output Behavior]

The response should first briefly identify the matched person and summarize the relevant evidence, and then provide the final answer grounded only in the corresponding context.

Table S.17: Judge prompt used for answer correctness.

Evaluation Judge Prompt:

You are an evaluation judge for a question-answering benchmark. You will be given a gold answer and a model prediction, and must decide whether the prediction is correct.

[Judging Rules]

1. For factual answers (*non-abstain*), the prediction is CORRECT if its final answer conveys the same core information as the gold answer, even if the wording differs.
 - Minor paraphrasing is acceptable.
 - The prediction is also CORRECT if the gold answer is explicitly contained in the prediction as a clear and stated fact.
2. The model receives multiple memory contexts and must identify the correct person before answering.
 - If the prediction lists multiple conflicting answers across different people, it is WRONG.
 - If the gold answer is only mentioned in passing while a different answer is presented as the model’s conclusion, it is WRONG.
 - If the model hedges without committing to a final answer, it is WRONG.
3. For abstain gold answers (“I cannot determine...”), the prediction is CORRECT only if it also abstains.
 - Any concrete answer in this case is WRONG.
4. Output exactly one word: CORRECT or WRONG.
 - Do not provide any explanation.

[User Input Template]

Input:

Gold answer: {gold}

Model prediction: {pred}

Verdict:

Table S.18: Visualization of the structured prompt used for automated persona profiling and attribute enrichment.

Personal Profile Builder Prompt (JSON-only):

You are a personal profile builder for a multimodal personalization benchmark.

[Task]

Your job has two stages:

1. **EXTRACT**: pull out any facts explicitly stated in the source text (e.g., name, job, MBTI).
2. **ENRICH**: for every field still null after extraction, invent a plausible and realistic value consistent with the dialogue style.

Think like a person writing down memory notes: assign believable personal attributes such as a hobby, a past travel experience, a job, an MBTI type, a memorable quote, an emotional state, or a physical appearance so that the profile feels like a real person someone might remember.

[Constraints]

1. Never contradict explicitly stated facts in the source text.
2. Keep each value concise (1–2 sentences). Values for experience/preference fields should reflect realistic, personal anecdotes.
3. Output must be strict JSON matching the profile schema.

[Input]

concept_id: {concept_id} source_type: {dialogue_type} SOURCE:
{source_text}

[Profile Schema]

```
{
  "hobby": "<string>",
  "major": "<string|null>",
  "affiliation": "<string|null>",
  "role": "<string>",
  "interest": "<string>",
  "topic": "<string>",
  "quote": "<string>",
  "location": "<string>",
  "environment": "<string>",
  "emotion": "<string>",
  "appearance": "<string>",
  "clothing": "<string>",
  "hairstyle": "<string>",
  "voice_identity": "<string|null>",
  "source_style": "<string>",
  "travel_experience": "<string|null>",
  "food_preference": "<string|null>",
  "life_event": "<string|null>"
}
```